

Tawhid of the Zeros

The Riemann Hypothesis as Qadar at One-Half, Determinate and Unread

Mohammad F. Islam, PhD¹

¹Independent Researcher, United States. Correspondence: islammm@alumni.iu.edu

The Riemann Hypothesis is a determinate arithmetical claim, and this essay asks not whether it is true but what kind of statement it is and what kind of openness it carries. The critical line is read as a decreed center, qadar, the fixed locus of the reflection composed from the functional equation and conjugation, elected before any zero is placed and elected by nothing in the plane it is drawn upon, which offers a continuum of equal candidates. The hypothesis is then the conjecture that the primes keep faith with that center of their own multiplicative nature, fitra, the many resolving as one, tawhid. The essay's central claim concerns the openness itself. The hypothesis is a determinate truth the primes already hold, written and fixed, and veiled from every finite instrument by eleven counted obstructions, of which three are proven non-entailments and the last is of a different order from the other ten. The fold wall: the reflection axioms alone do not entail the conclusion, since the extended Selberg class of degree one contains the Davenport-Heilbronn function, which carries the fold at one half and places infinitely many zeros off it. The counting wall: an Euler product with prime counting of the classical grade does not entail it either, since the Diamond-Montgomery-Vorhauer generalized-prime system carries both and sends its zeros to a de la Vallee Poussin curve at the abscissa one, with no fold anywhere in the picture. The finite-verification wall: no mountain of instances reaches a universal over the integers, and three such mountains have collapsed. The first two walls are not nested and their conjunction is the essay's central mathematical arrangement: each axiom set has its own countermodel, the two models fail in orthogonal directions, one transverse to the center and one longitudinal toward the edge, and the zeta function is the single object carrying both the fold and the Euler product over the primes of the integers, so no consequence of either axiom set alone decides the question, which is theorem, while the corridor that reading opens is structural and its one assumption is named where it is drawn. Each wall is a limit on the reader and never a permission granted to the zeros, and reading a witness function's straying as a freedom exercised by the zeta zeros is a leak from an instrument fact to an object fact, refused here at the outset. Two errors are then refused with one symmetric discipline: the manufactured room that reads a veil as ontological openness, and the premature seal that reads a located witness as a delivered proof. The honest posture is tawakkul. The essay proves nothing, adds no mathematics, and every mathematical premise is a classical result of others. Theology does no mathematical work in it, in either direction.

KEYWORDS Riemann Hypothesis, theology of mathematics, qadar, tawhid, fitra, tawakkul, ghayb, Davenport-Heilbronn function, Beurling generalized primes, Euler product, von Mangoldt correlations, truth and provability

The question and the frame

The Riemann Hypothesis asserts that every nontrivial zero of the zeta function lies on the vertical line at real part one half. It is a determinate arithmetical claim with a definite truth value, open since Riemann's memoir of 1859.¹ The literature on it is a literature about its truth. This essay sets that question aside and asks a different one.

Granting the hypothesis a definite answer not in hand, what kind of openness is that.

The answer given here is single and it is structural. The openness is not a mystery to inhabit. It is a veil over a written thing. The truth of the hypothesis is a determinate fact the primes already fix, and it is hidden from us behind eleven counted obstructions, of which three are non-entailments, each a theorem, each a limit on the instrument and none a softness in the object. The first is a countermodel inside

the reflection axioms. The second is a countermodel inside the Euler-product-and-counting axioms, and the two are not nested. The third is the wall that no finite verification of a statement about all integers can cross. The decree is written. We cannot read it. Those are two facts and the entire discipline of the essay consists in keeping them apart.

A separation is imposed at the outset and held throughout, because the reading stands or falls on it. A fact about what an instrument can resolve is not a fact about what the object contains. The counterexample functions that appear below do not stray. They show that a particular reading channel returns the same verdict whether or not straying occurs, which is a statement about the channel. To convert that into a claim that the zeros of the zeta function possess and decline an alternative is to carry an instrument fact into the object register with no bridge between them, and no such bridge is offered here or exists. That conversion is the one move this essay refuses first, before it makes any other.

The theology is bound to the same separation. Everything theological here reads the relation between a center that is fixed and a truth about that center that is written and unread, offered as interpretation of a settled structure, never as a result and never as evidence for or against the hypothesis. Theology does no mathematical work here, and the doing-no-work is symmetric, which is the condition of its honesty and is stated plainly in its own section. Section two establishes the settled mathematics. Section three states what the two counterexamples do and do not establish. Sections four and five read the decree and the innate nature. Section six sets the two counterexamples out as a lattice and draws the joint necessity from it. Section seven reads the unity. Section eight raises the veils and counts them. Sections nine through eleven read the marginal line, the partition, and the silent center. Section twelve refuses the two errors. Section thirteen fixes the ledger. Section fourteen answers objections, section fifteen states exactly what the reading delivers and what it does not, and section sixteen concludes.

The settled mathematics

The geometry first. because it is elementary and because it is the entire content the symmetry supplies. Write a complex number as s with real part σ and imaginary part t . The functional equation pairs s with one minus s , and Schwarz reflection pairs s with its conjugate. Their composition is the reflection sending σ plus it to one minus σ plus it , a reflection of the strip across the vertical line at σ equal to one half, fixing the imaginary part. The fixed set of that reflection is exactly the line at one half. The critical line is therefore not defined by where the zeros happen to lie. It is the mirror of the functional equation, fixed before any zero is located.² The signed horizontal distance of a point from the line, the quantity σ minus one half, is the coordinate the reflection negates, the odd coordinate of the symmetry, and the hypothesis is the assertion that this coordinate vanishes at every nontrivial zero.

The decisive fact is that the fold is shared, and the exact sense in which it is shared has to be stated precisely, because the loose version collides with a theorem. Hamburger proved in 1921 that a Dirichlet series

satisfying the zeta functional equation on the nose, the same gamma factor at the same conductor, is a constant multiple of the zeta function.³ Nothing else does archimedeanly what zeta does. A counterexample therefore cannot match zeta in that data and must not be said to. What it matches is the fold. **And the exhibited witness differs in both of the data Hamburger requires, not one.** Its completion holds at a conductor of five and with the odd archimedean factor, and it fails every even-parity completion at any conductor, while the object holds at level one with the even factor and fails the witness's form. **So the separating datum is the level-one completing factor, conductor and archimedean parity together, with the growth class,** and naming the archimedean factor alone understates the distance. **The fold-sharing is thereby exact rather than approximate:** the witness shares the reflection and shares nothing of the completion, which is the strongest form the argument takes. In 1936 Davenport and Heilbronn exhibited a Dirichlet series satisfying a functional equation of Riemann type across the same reflection axis at one half, carrying the odd-character gamma factor at conductor five in place of zeta's even factor at conductor one, and proved that it has zeros off that axis.⁴ Both functions lie in the extended Selberg class of degree one, the axiom class of Dirichlet series carrying a Riemann-type functional equation across the axis at one half with the gamma factor and the conductor left open,⁵ and it is membership in that class, and not identity of archimedean data, that the two share. The distinction is not pedantry. It is the whole content of the wall, because what the two functions have in common is exactly the involution and its fixed locus, and everything else about the completion is a parameter the class leaves free.

Later work sharpened the picture. The Davenport-Heilbronn function has infinitely many zeros off the axis, the count up to height T growing in proportion to T , and infinitely many on the axis as well, so it is not pathological everywhere.^{6,7} The off-line zeros are explicitly computable. One sits at real part 0.808517182456637, an offset of 0.308517182457 from the axis, at height 85.699348485378.⁸ Its functional-equation partner, one minus that value, is an equal zero at the opposite height. The partner standing at the same height on the other side of the axis is a different object: it is the image under the composed reflection, one minus the conjugate, which is the involution this section opened by defining. The two partners are not the same point and the composition is what supplies the mirrored family at fixed height. The distinction was drawn three paragraphs ago and it is kept here.

Box 1 | The two completions, checked

The distinction the paragraph above turns on is mechanical and was re-executed rather than taken on report. Write the Davenport-Heilbronn coefficients as the sequence one, tau, minus tau, minus one, zero repeating with period five, where tau is the positive root of tau squared plus one plus root five times tau minus one, numerically 0.2840790438404123. The root is cross-checked against a second closed form, the difference of the square root of ten minus twice the square root of five and the number two, divided by the square root of five less one, which agrees with the quadratic root to forty digits. Complete the series with the odd-character factor at conductor five and the ratio of the completed function at s to its value at one minus s returns one to fifty digits at each of three test points off the axis, taken at s equal to 0.3 plus $7.1i$, 2.4 minus $3.3i$, and 0.9 plus $21.0i$. Complete the same series with the zeta factor at conductor one and the same ratio at the same three points returns 1.2533 minus $0.5770i$, 0.0388 plus $0.0265i$, and minus 0.3616 plus $0.3810i$. The zeta functional equation fails outright. Two further checks confirm the reading. The ratio of the Davenport-Heilbronn function to the zeta function at the integers two through five is 0.60797 , 0.84314 , 0.93416 , 0.97100 , not constant, so the function is not a multiple of zeta and Hamburger's theorem is not violated. And the function vanishes at the odd negative integers while taking the values minus 0.9136 and 6.8972 at minus two and minus four, the trivial-zero signature of an odd character and not of zeta. Warrant: arithmetic verification of a cited classical result at fifty digits; no new mathematics and no new claim.

The consequence is model-theoretic and it is immediate. Every property entailed by the class axioms alone holds of every member of the class. The conclusion that all nontrivial zeros lie on the axis is false of one member. Therefore the class axioms do not entail the conclusion. One exhibited model closes the question and no further argument is needed. No refinement of reasoning drawing on those axioms alone can recover the entailment either, since a refinement is still a consequence of the same axioms and would still have to hold of the model where the conclusion fails. This is an insufficiency of one tool, stated at theorem grade, and it is not an impossibility of the goal.

The boundary of what the symmetry leaves out can be drawn sharply, and drawing it names the missing structure. What the zeta function has and the Davenport-Heilbronn function lacks is the Euler product, the factorization of the Dirichlet series into a product over primes, the analytic image of the multiplicativity of the integers. The Davenport-Heilbronn coefficients are not completely multiplicative and the function admits no Euler product. The two differ in other respects that do not bear on the location of the zeros, the zeta function carrying a pole at one while the counterexample is entire, but at the level that governs the offset the Euler product is the difference the localization turns on here.

Box 2 | The localization, and to whom it belongs

That the hypothesis is decided on the multiplicative axis rather than by the functional-equation symmetry is not a result of this essay. It is the organizing principle of the Selberg class, the axiomatic family of Dirichlet series defined by a functional equation, a Ramanujan bound, and an Euler product, for which the analogue of the hypothesis is conjectured.⁹ The Euler product is a separate axiom, and the Davenport-Heilbronn function is the standard object showing it cannot be dropped. The expository literature states it plainly: a proof would need the functional equation, the Dirichlet series, and the Euler product together, precisely because counterexamples show the first two without the third permit zeros off the line. A degree-one necessity theorem of Kaczorowski and Kulas sharpens the direction: a degree-one member of the extended Selberg class meeting the hypothesis or a nontrivial density estimate must carry an Euler product, so arithmetic structure is necessary.¹⁰ This essay takes the localization as settled and given, and its own reading begins at the interpretation of that settled fact. Warrant: theorem-grade as cited, and the essay adds it nothing.

A second insufficiency stands one channel over, and it is the sharper of the two because it holds against an instrument that already carries the multiplicative form. Beurling generalized number systems replace the primes by an arbitrary discrete multiplicative generating set and build the zeta function of that system as an Euler product over those generalized primes. Diamond, Montgomery, and Vorhauer constructed such a system whose integer counting is asymptotically regular and whose prime counting is as irregular as the classical zero-free region permits, an omega estimate and not an upper bound, which is what the phrase large oscillation in their title names. The zeros of its zeta function lie on a curve of the de la Vallée Poussin shape, one minus a constant over the logarithm of the height, with none to the right of it, so they accumulate toward the abscissa at one.¹¹

Two things must be said exactly about this witness, because the loose reading overstates it in one direction and understates it in the other. It carries no functional equation. Beurling systems generically have none and this construction has none, so it has no fold, no fixed locus, and no mirrored families about any axis, and its zeros are not scattered about the line at one half. Understated, and this is the part that matters: that is a stronger failure than straying across a fold, not a weaker one. An Euler product together with prime counting of the classical grade does not place the zeros usefully at all. It does not put them at one half, it does not keep them left of one half, and it does not keep them off the boundary. The counting channel is not merely blind to the offset. It does not constrain the offset.

Two further classical facts complete the picture. A positive part of the hypothesis is already a theorem. Hardy proved in 1914 that infinitely many zeros lie on the line.¹² Conrey proved in 1989 that more than two fifths of all the zeros lie on it.¹³ The on-line configuration is not a speculative ideal. It is provably occupied, by infinitely many zeros and

by a positive proportion of them. The hypothesis is the claim that the occupation is total.

The record of the attempts

The hypothesis has been worked on continuously since it was stated, and the record is not one of neglect. It is a record of **partial results that accumulate without converging**, and the shape of that non-convergence is what the rest of this reading is about. *The attributions and dates below are recalled and stand at their published grades.*

It begins as a remark. Riemann states it in 1859 inside a memoir about counting primes, and does not dwell on it. It is not presented as a problem. It is presented as something evidently so.

The first real ground is taken by pushing the zeros off a boundary. Hadamard and de la Vallée Poussin, independently in 1896³³, show no zero sits on the line where the counting would break, and the prime number theorem follows. **The method locates zeros by excluding a region**, and the region it can exclude never narrows to the axis itself.

Then the axis is shown to be populated, and populated richly. Hardy in 1914 proves infinitely many zeros lie on it. Hardy and Littlewood in 1921 raise this to a positive proportion. Selberg in 1942 obtains a positive density. Levinson in 1974 reaches at least a third³⁴; Conrey in 1989 at least two fifths.

Read the sequence carefully, because it is the essay's first exhibit. Infinitude, then proportion, then a third, then two fifths. **Every step is real and the sequence does not approach one.** The methods raise a lower bound on how many zeros are on the axis, and a lower bound on a proportion does not become a universal by improving. **The road is genuinely open and it does not lead there.**

Meanwhile the analogue is proved outright. Weil in 1948 establishes the corresponding statement over function fields, and Deligne extends the surrounding conjectures in 1973. **This is the one lift on the record**, and it matters enormously to what follows: it shows the statement's shape is provable somewhere. The instrument was a general theorem from geometry whose specialisation gave what was needed.

The statistical structure is uncovered. Montgomery in 1972 finds the pair correlation of the zeros matching a law from random matrices, and Odlyzko's computations through the 1980s confirm it to great height. **The zeros turn out to be beautifully regular, and regularity of spacing says nothing about location.**

And the whole question is compressed into a single number. De Bruijn and Newman, across 1950 and 1976, deform the function by a heat flow and show that a single real constant governs whether all zeros are real. **The hypothesis becomes one equality.** Rodgers and Tao prove one side of it in 2018, from below. A collaborative effort pushes the other side to roughly a fifth. **One half of the pincer is a theorem and the other is a finite computation that must be repeated forever.**

Alongside, the coefficient route. Csordas, Norfolk and Varga prove the second-order inequalities in 1986³⁵; Griffin, Ono, Rolen and Zagier prove in 2019³⁶ that the higher-degree conditions hold for every degree, past a threshold that depends on the degree. **The proven region is complete except in one corner**, and the threshold is not uniform.

Where each attempt stopped, and why the places differ

Set the stopping points side by side and they do not form a queue behind one obstacle. **They are distinct, and each has its own mechanism.**

The zero-free region method stops at a shape. It excludes zeros from regions that narrow toward the axis and never reach it. The obstruction is not difficulty but form: **the method yields a margin, and the hypothesis asks for a line.** A margin is not a small line.

The proportion methods stop at a limit that is not one. Each improvement is genuine and each is a lower bound. **No sequence of lower bounds on a proportion reaches a universal**, and this is a fact about the kind of statement being proved rather than about the ingenuity applied.

Finite verification stops at every height it reaches. Zeros have been checked past any number a person would name. **The hypothesis quantifies over all of them**, and a universal over an infinite domain is not approached by exhausting a finite part of it. Every verification is complete and none is progress toward completion.

The reformulations stop at their own reflection. Seven published criteria state the hypothesis in other vocabularies, and each is **equivalent** to it. A criterion equivalent to what it would decide is a translation. **It touches every part of the target and therefore leaves no part untouched to gain purchase on**, which is why the sequence of restatements has produced no lift in over a century.

The geometric route stops at an absent object. What carried the analogue was a surface, obtained by crossing a curve with itself over a base beneath it. **Over the integers there is nothing beneath**: the corresponding product collapses back to a curve, and the surface an intersection theorem would stand on is not there to be stood on.

The two live routes stop at a missing uniformity. The flow route needs a constant that does not degrade as its parameter shrinks; the coefficient route needs a threshold that does not grow with the degree. **Neither is in hand, and both are the same species of gap read on different variables.**

And the analogue's success stops at the boundary of its setting. It was proved where a base existed beneath the object. **The proof is not defective and does not transfer**, because the thing it stood on is absent here.

The blind instrument

Read the two counterexample functions correctly and they say one thing, exactly, and it is a statement about reading and not about zeros. Each exhibits a model in which a set of axioms holds and the conclusion those axioms are asked to deliver is false. That is the entire mechanism, it is two lines long, and it needs nothing else.

It is worth saying why the tempting shortcut is not the mechanism, because the shortcut is common and it does not work. The signed distance from the axis is the odd coordinate of the involution, the quantity the reflection negates, and one is inclined to argue that every instrument built from the reflection is even in that coordinate and so cannot read it. Both halves of that argument fail. The premise is false: the completed zeta function satisfies its functional equation and the composed reflection sends it to its own conjugate, so its real part is even in the offset while its imaginary part is odd, and the imaginary part is a functional of exactly the reflection data. And the conclusion would not follow even from a true premise, because the hypothesis does not ask after the sign of the offset. It asks whether the offset vanishes, and vanishing is even. An argument that the reflection cannot see an odd quantity is aimed past the target, since the target is invariant.

The parity fact is real and it has a different address. What no invariant of the reflection can carry is the direction of a violation, which side of the axis a straying zero fell to, because that is the odd datum and every invariant discards it. So parity governs the sign of an offset and never its existence, and any instrument that squares away the sign, as every quadratic form built from the reflection does, has discarded exactly the information the sign question needs and none of the information the hypothesis needs. Two questions, two addresses, and conflating them is what makes the shortcut attractive.

Two further facts about the geometry sharpen why the reflection is helpless here, and both are elementary. The direction of departure is **one-dimensional**: the composite involution fixes one coordinate and reverses exactly one, so leaving the axis is a single degree of freedom and everything else a zero can do, its height, is motion **along** the axis and not away from it. A departure therefore carries a sign and a distance and nothing further, and there is no third way to be off. And the departures come in **mirror families**. If a zero strays, the fold and the conjugation together produce its partners on the far side at the same height, so a strayed configuration is **symmetric about the axis** rather than lopsided across it. That is why an off-axis population **satisfies** the fold rather than violating it, which is exactly what the countermodel exhibits and what makes the countermodel possible at all. The reflection sees the family and never the member, and asking it which side a zero fell to is asking a symmetric object to name an asymmetry it was built to average away.

The exact strength of the claim must be stated and not exceeded, because a great deal of loose reading gathers at precisely this point. The claim is about entailment. Relative to the class axioms the conclusion is not forced, and the counterexample is a member of the class where it fails. The claim is not that the zeros of the zeta function

are free to stray, which the counterexample does not establish and which no fact about a different function could establish. The counterexample is not an arithmetic object at all. It has no primes, no factorization, no counting function, no population. It is a tuned linear combination built to carry a symmetry and nothing else, and the only thing that actuates when its off-line zero is located is the computation that locates it. Its wandering is the wandering of an object with no arithmetic to be faithful to, which is why it is a perfect witness to the blindness of the symmetric channel and no witness at all to anything the primes do.

The Beurling systems make the point harder to evade, because they are arithmetic objects, and they say the same kind of thing one channel over. They carry generalized primes and an Euler product over those primes, their integer counting is regular, and their zeros run to the abscissa. What that establishes is that the counting channel, like the fold channel, does not force the conclusion. It establishes nothing about the primes of the integers, which are not the generalized primes of the construction, and the running belongs to the system that runs.

So the correct reading of both witnesses is the same and it is negative throughout. Neither exhibits a permission. Each exhibits a set of axioms with a model in which the conclusion fails, which is what it means for a channel to decide nothing. The temptation is to promote the second into the first, to hear "the symmetry does not forbid straying" as "straying is permitted" and then as "the zeros hold an alternative they decline." Every step of that promotion is a category error, and the last one is the worst, because it takes a property of a reading channel and dresses it as a modal property of the integers. The zeros of the zeta function do exactly one thing, whatever it is, fixed by the multiplicative structure before any reading begins. Nothing about a blind channel adds an alternative to that.

What survives, and it is considerable, is non-compulsion. The symmetry compels nothing. It fixes a mirror and issues no order about occupancy, and the counterexamples prove that the absence of the order is real rather than merely unproven. This is the whole of what the witnesses deliver and it is exactly what the reading below requires. A decree that compels nothing does not need a realized defector in order to be non-compulsive.

The origin is present

The world has a beginning in God . The origin is not a receded past event that the world has since travelled away from. It is present now, in the form the world holds. **It is the line every existing thing stands upon**, and standing on it is not a position a thing occupies but the condition of its being a thing at all.

The consequence, which follows by definition

From the absolute oneness of the Source, one thing follows, and it follows by definition rather than by argument. **Nothing falls outside that line.**

Run the supposition. Suppose a form did fall outside it. **That form would then require a source the first does not account for.** A

second source would have to stand beside the first, generating what the first did not. **There is no second. That is what *one* means.**

So the supposition denies the premise it was made under. **What looked like a possibility narrowed down was never a possibility.** The line is not where things happen to be found. **It is what having a source at all consists in.**

Qadar at one-half: the decreed center

The word qadar carries, before any later elaboration, the plain sense of a measure that is set, a determination laid down. Read the critical line in that plain sense and the fit is exact. The line is the measure of the zeros, the value their offset is to take, and it is fixed before any zero is placed, because it is the mirror of the functional equation and not a property read off the zeros after the fact. The center is decreed. It does not wait on the zeros to discover where it is.

There is a sharper sense in which the center is set rather than found, and it is geometric. The plane on which the zeros lie privileges no line over any other. It is even throughout, and for any vertical line there is a reflection holding it fixed, each such reflection the equal of every other. Nothing in the geometry elects the line at one half. What elects it is the functional equation, whose fixed line falls exactly there, and that reflection descends from Poisson and Jacobi through Riemann, not from the plane it is drawn on. So the decreed center is founded on nothing in the plane itself. It is chosen out of a continuum of equal candidates and laid upon a ground indifferent to it, a measure set upon what does not itself contain it, owing its place to the decree alone.

The point sharpens once more when the rest of the picture is examined for what it owes to the draftsman, and the sharpening requires a declaration no invariance claim can do without. An invariance claim is empty until the acting group is named, since a magnitude is structural at one register and orbit at the next. Two registers are in play here and both should be stated. At the analytic register, where the acting group is the affine maps commuting with the involution, the axis is carried to itself and the address one half is a fixed point identically, while the strip width still scales freely. At the bare incidence register, where the acting group is the homeomorphisms of the strip, the width and the address are both orbit and what survives is only incidence, whether a point coincides with its own image. The claim under audit is an incidence claim, whether a zero sits on its own mirror, so the incidence register is the adjudicating one and the reading below is taken there.

The width of the critical strip, the address of the line within it, its position at the middle, and the numerical value one half are readings taken off a chart. Re-coordinate the plane and each of them moves while the structure stands unchanged, since a monotone change of the exponent variable carries the strip to a strip of any width one likes and the line to any address inside it, the zeros and their relations riding along untouched. What does not move is the fold. The reflection is elected by the functional equation, the fixed set of an involution is an invariant of that involution and of nothing else, and the coincidence of a point with its own mirror image survives every chart. So the

decree is precisely the residue that remains when every manufactured magnitude has been stripped away, and the famous number is a label on the residue rather than the residue itself. Qadar is the fold, and the half is what the fold is called in the coordinates we happen to use.

Set the decreed center beside the non-compulsion of section three and the shape of the hypothesis appears. The center is decreed and nothing compels the zeros to it, and the hypothesis is the proposition that they rest exactly there. The decree is not a force. A force would be an efficient cause, a push that pins the zeros and forbids them to leave, and there is no such cause at the level of the symmetry, since the very same symmetry present in a function whose zeros are elsewhere pins nothing. To read the line as a force is to read it as compulsion, and compulsion is what the witnesses rule out. The measure is set, the compulsion is absent, and the conjecture is that the primes come home to the measure of their own nature.

Fitra: the multiplicativity of the integers

If nothing compels the zeros to the line and yet, by the hypothesis, they arrive there, the arrival must be accounted for by what the zeta function is rather than by anything done to its zeros. The word fitra names the primordial nature a thing is made with, the disposition it returns to when nothing forces it otherwise, and the question is exactly what that nature is here.

The obvious answer is multiplicativity, and the Beurling systems show that the obvious answer is too coarse. Those systems are multiplicative. They carry an Euler product over their own generalized primes, they satisfy the prime-number theorem with a classical error term, and their zeros stray. Whatever holds the zeros of the zeta function to the line, if they are held, is therefore not multiplicativity as a general form. It is the multiplicativity of the integers in particular: the specific population, its discreteness, its integrality, the fact that the generating set is the primes of the natural numbers and not an arbitrary discrete set arranged to imitate them. Fitra is not a form the object shares with its imitations. It is the nature of this population.

The localization sharpens once more when one asks where the deciding content actually sits, and here the mathematics is precise. Several classical reformulations reduce the hypothesis to the positivity of a single arithmetic object built from the von Mangoldt function, the function that weights each prime power by the logarithm of its prime. The self-contribution of that object, its diagonal, is fixed by unique factorization and is unconditionally well behaved. What is not fixed by factorization is the off-diagonal, the correlation of the von Mangoldt weights with their own shifts, the sums of a prime-power weight against the weight a fixed distance away. Deterministic control of exactly these shifted correlations is the frontier of Chowla and Hardy-Littlewood and is open in the correlated cases.¹⁴ That it is harder than the hypothesis is the field's expectation and not a theorem, and it is carried here at that grade. The fundamental theorem of arithmetic governs the diagonal and says nothing about the off-diagonal. So the innate nature reaches each prime's own contribution and does not, of itself, reach the correlations.

That the correlations are where the hypothesis is decided is a structural reading and is graded as one throughout this essay, at the grade section thirteen states and defends. It is not a theorem that the hypothesis is equivalent to deterministic control of the shifted correlations, and no such equivalence is claimed. What is claimed is that every road which would imply the hypothesis rather than restate it passes through that control when it is followed to where it binds, which is a map of the roads and not a theorem about the destination.

This is not a defect in the reading. It is the reading made exact, and it accounts for both counterexamples without straining either. The Davenport-Heilbronn function carries the fold and no Euler product, so it has a center and no nature, and its zeros leave the center. The Beurling systems carry an Euler product and no fold, so they have a nature and no center, and their zeros run to the edge. Neither carries both. The zeta function is the object that carries both, and section six sets the four cases out as the lattice they form.

The Aristotelian analysis. of a single zero on the line makes the structure precise and locates the error of the force-first tradition.¹⁵ The material cause is the primes, through the coefficients of the series. The formal cause is the multiplicative nature of the integers, the fitra. The efficient cause is the constraint the force-first programs seek, and the counterexamples show it is not supplied by the symmetry, which is a statement about supply and never an established absence. The final cause is the decreed center. The coincidence of the formal and the final cause is the heart of it. For a natural thing the end is the complete realization of its form, so form and that for the sake of which name one thing under two descriptions. The multiplicative nature of the integers is what the zeta function is, and it is also, on the hypothesis,

that in virtue of which the zeros rest on the line. Submission to that nature is not coercion. It is the way of the thing.

The error of the force-first tradition can now be named without polemic. It is a category mistake, the search for an efficient cause where the operative cause is the innate form. The tradition asks what forces the zeros onto the line and looks for a symmetric structure that does the forcing. But the symmetry is the wrong category of cause, shared with a function whose zeros are elsewhere. This is not a discouragement of the spectral program but a clarification of what it must do. A spectral object that decides the hypothesis cannot merely produce the symmetry, the reality of a spectrum, which is shared with the counterexample. It must encode the multiplicative nature of the integers into its very construction, and in particular it must reach the correlations, which is the standing difficulty of the program and the reason a natural operator has been so hard to find. The easy part of the target is not the deciding part.

The lattice of the two witnesses

The two counterexamples are not two stages of one progression, and it is worth being exact about the arrangement they do form, because it is the sharper one and it carries the whole architecture of the problem.

Two axiom sets are in play. Call the first the fold: a Dirichlet series carrying a Riemann-type functional equation across the axis at one half, gamma factor and conductor left open, which is the extended Selberg class of degree one. Call the second the nature: an Euler product over a discrete multiplicative generating set, with prime counting of the classical grade. Each set has a model in which the conclusion fails, and the two models fail in different directions.

Table 1 The lattice. Each witness kills one entailment, and they are not nested.			
Axioms carried	Model	Where the zeros go	Conclusion
Fold only, no Euler product	Davenport-Heilbronn	off the axis, mirror-paired at fixed height, bounded in the strip	false
Euler product only, no fold	Diamond-Montgomery-Vorhauer	onto a de la Vallee Poussin curve, accumulating at the abscissa one	false
Fold and Euler product	the zeta function, and the Selberg class	conjectured on the axis	open
Neither	a generic Dirichlet series	unconstrained	no content

Note: The two failures are not the same failure weakened or strengthened. Each is a separate non-entailment established by a separate exhibited model. Warrant: theorem-grade per row as cited; the lattice arrangement is structural, and it is the essay's arrangement of results belonging to others.

Read the lattice and two statements come off it, and they carry different grades. The first is theorem and it is exact: no consequence of the fold axioms alone decides the question, by the first model, and no consequence of the Euler-product-and-counting axioms alone decides it, by the second. This is closed under refinement, which is what makes it a wall rather than a difficulty: a strengthening of fold-only reasoning is still a consequence of the fold axioms and must still hold in the model where they hold and the

conclusion fails, and the same for the nature. Neither axiom set is a candidate, at theorem grade, now and permanently.

The second statement is the corridor, and it is structural. Read the two non-entailments together and one is inclined to say that a proof must therefore use both. That does not follow from the two models and the gap should be named rather than glossed. Two countermodels establish that each of two resource sets fails alone. They establish nothing about a third resource reducible to neither, and nothing above forecloses one. This essay names such a candidate itself, in its thirteenth section: deterministic control of the correlations of the primes with their own shifts is not a consequence of the fold axioms and is not a consequence of counting control of the classical grade, and if the question is decided there it is decided by an ingredient the

lattice does not chart. So the corridor is a reading of where the roads run and not a theorem about which road must be taken, and it is carried at that grade throughout.

The inflation the split forecloses is worth naming, because it has a shape and the shape recurs. To pass from each alone fails to both are required is to assume the resource space is exhausted by the two sets in hand, and that assumption is supplied by nothing. It is the same sign-flip one level up from the fence the fold wall already carries: the insufficiency of one tool is not the impossibility of the goal, and the joint insufficiency of two tools is not the exhaustion of the toolbox.

The two failure directions are geometrically distinct, and the distinction is the reason there are two walls. The Davenport-Heilbronn zeros fail transversally. They leave the axis along the odd coordinate of the involution, stay mirror-paired at fixed height, and remain bounded inside the strip, so the fold is present and the occupancy is wrong. The Diamond-Montgomery-Vorhauer zeros fail longitudinally. They run along the strip toward the abscissa at one, the edge of the half plane where the Euler product converges absolutely, and there is no fold anywhere for them to be on the wrong side of. One failure is across the center. The other is toward the wall.

That second direction is worth one more sentence, because it says what the Euler product by itself reaches. Read the multiplicative structure thermodynamically, as a gas whose modes are the primes and whose partition function is the Dirichlet series, and the abscissa at one is exactly where that gas diverges, the pole of the zeta function and the boundary of absolute convergence. The multiplicative content by itself terminates there. It reaches the edge at one and no further, and the Beurling witness is the demonstration: strip the fold, keep the gas, and the zeros run to precisely that edge. The value one half is not in the multiplicative content at all. It is the fixed set of the involution, and it enters only with the fold.

So the honest statement of what separates the zeta function from its two nearest imitators is a conjunction and not a single feature. It carries the fold, which is the center, and it carries an Euler product over the primes of the integers in particular, which is the nature. Each imitator drops one term of the conjunction and each fails, in the direction the dropped term was holding. The hypothesis is the claim about the conjunction, and no reading of it that leans on either term alone has been licensed by anything above.

Two halves of the picture carry different warrant, and the honesty of the arrangement is in keeping them apart. That the two witnesses fail is theorem. That the zeta function does not is conjecture. And neither witness is evidence about the zeta function in either direction, because they are different objects and the difference is exactly the feature at issue. A counterexample that isolates the insufficiency of a tool tells you about the tool.

Tawhid of the zeros

Tawhid is the affirmation of oneness, and the hypothesis is, in its structure, an affirmation of oneness at three levels the mathematics makes exact. Two of them concern the zeros and the third concerns

the object that carries them, and the third is the one the lattice of the previous section supplies.

The first is the oneness of each zero with its own reflection. The critical line is the fixed set of the reflection that sends the offset to its negative, so a point on the line is a point identical to its own mirror image. A zero on the line is a zero made one with its own reflection, no longer one of a mirrored pair standing apart but a single thing coincident with its image. Off the line a zero and its mirror are two. On the line they are one.

The second level is the oneness of the many upon the single center, and here the localization of section five does its work. The nontrivial zeros are an infinite multitude. The hypothesis is the claim that this multitude lies entirely upon the one line, and the deciding structure is not the fidelity of each zero taken alone. It is the fidelity of the whole correlation, the collective behavior of the multiplicative weights against their own shifts, resolving onto the line. The unity the hypothesis asserts is achieved from within, by the integrity of a common nature realized across the multitude, and not imposed from without by a constraint, since no constraint is supplied. A unity imposed by force would be a cage holding together what would otherwise fly apart. The unity here is the other kind, the many coming to one because their shared nature draws the whole correlation to the one center. The decree set the measure, the innate nature is the way the measure is met and the correlations are where it is met, and the unity is what meeting it looks like when the many meet it together as one correlation.

The third level is the oneness of the two terms in one object, and it is the level the lattice makes visible. The decree and the nature are separable, and the two counterexamples separate them. The Davenport-Heilbronn function holds the fold and no Euler product: a center with no nature to keep it, and its zeros leave the center. The Diamond-Montgomery-Vorhauer system holds an Euler product and no fold: a nature with no center to keep, and its zeros run to the edge. Each is a partition of what the zeta function holds together, and each partition fails in the direction of the term it dropped. The zeta function is the object in which the two are not two. That is the third oneness, and unlike the first two it is a settled fact about the object rather than a conjecture about its zeros: the conjunction is exhibited, whatever the zeros turn out to do.

The three levels stack in the order the essay has been reading them and the stacking is the whole reading. The center is one, the fold, elected before any zero. The object is one, the fold and the nature met in a single function, which the lattice exhibits and the counterexamples confirm by failing when either term is removed. And the multitude, on the hypothesis and not otherwise, is one upon the center. The first two are settled. The third is the string held at terminal suspension, determinate and unread, and the title of this essay names it as the structure under examination and never as a result in hand. To read the third as delivered because the first two are settled would be the premature seal of section twelve committed on the cover, and it is refused there as it is refused here.

Table 2 | The three terms and their mathematical residence.

Term	Mathematical residence	What it means
Qadar, the decree	the fold, the fixed locus of the reflection, invariant under every chart	a measure set before any zero, elected by the functional equation and by nothing in the plane
Fitra, the innate nature	the multiplicativity of the integers in particular, reaching each prime and not, of itself, the correlations	the disposition of this population, distinguished from its imitators by the population and not by the form
Tawhid, the unity	the fold and the Euler product met in one object, and the whole von Mangoldt correlation resolving onto the one line	the two terms not two in the zeta function, which is settled, and the multitude at rest as one upon the center, which is not

Note: All three are premise-grade theological readings of theorem-grade mathematical structures. They are load-bearing on nothing in the mathematics.

The two names

And one of the twelve words carries the confusion already. The zeros the claim excludes are called trivial, and the ones it speaks of are called nontrivial. The names are honest bookkeeping and the mathematicians did nothing wrong by them: the first family falls out of the functional equation in a line, and the second is what remains after it.

But look at what the two families are. The trivial zeros are infinite and scattered across the whole negative axis. The nontrivial ones are claimed to sit on a single line. **By any structural measure the scattered family is the loose one,** and the disciplined family is the one whose name says it is the hard case. The word records **how cheaply a reader obtains them** and records nothing else, and a note about our effort has been read for a century and a half as a rank of being.

On the reading offered here, every zero is trivial. Each is determinate, placed, not negotiated, standing where it was put. **What differs is the veil, not the zero.** One family lifts its veil in a line and the other has not lifted in a hundred and sixty-seven years, and the seeing of the second must be earned where the seeing of the first is given. **Trivial and nontrivial name the price of the looking. They name nothing about the looked-at.**

The veils

The hypothesis is a determinate truth the primes already hold, and it is veiled from every finite instrument. Three of the veils are non-entailments and they are named first, because they are the structural core of the essay and each is a classical fact rather than a reading. Six

further obstructions sit beside them, counted and typed in the table below, and the ninth is of a different order from the other ten.

The fold wall

The fold alone does not decide, and the proof is one model. The extended Selberg class of degree one contains the Davenport-Heilbronn function, that function has infinitely many zeros off the axis, and therefore the class axioms do not entail that the zeros lie on it. Nothing further is needed and nothing weaker will do, since any argument drawing on those axioms alone is a consequence of them and must hold wherever they hold. The contrapositive is the sharpest sentence available on the problem. Any sound argument for the residence of the zeta zeros must carry a premise true of the zeta function and false of the counterexample, and the Euler product is exactly such a premise. This wall is permanent in the exact sense that a countermodel is permanent, not a difficulty to be out-worked by a finer argument of the same kind. And it relocates without vanishing when the hypothesis is dressed in a different costume. Restate the hypothesis in the language of the divisor sums, the Mobius function, or the counting error of the primes, and each restatement carries a barrier native to its own instrument. Change the clothing and the barrier moves to the new instrument. It does not disappear, because it was never a property of the clothing. Warrant: theorem-grade, with the witness cited.

The counting wall

The nature alone does not decide either, and again the proof is one model. The Diamond-Montgomery-Vorhauer system carries an Euler product over its generalized primes and integer counting of the classical grade, and its zeros accumulate on a de la Vallee Poussin curve at the abscissa one. So no argument whose premises are exhausted by an Euler product plus counting control of that grade can reach the conclusion. This wall is the harder of the two, because it stands against an instrument that already holds what the first wall showed to be missing, and because its failure is not a near miss. The zeros do not land slightly off the axis. They land as far from it as the construction permits, at the edge of convergence, with no fold in the picture at all.

Together the two walls bound the shape of any proof, in the split grade section six sets out. At theorem grade: no consequence of the fold axioms alone succeeds, and none of the Euler-product-and-counting axioms alone, and no refinement of either recovers the entailment. At structural grade, and only there: reading the two together suggests a proof draws on both and on more of the integers than their counting statistics carry, since counting of the classical grade is already inside the second model. That reading assumes the two sets exhaust the resources available, which nothing establishes, so it is a corridor and never a necessity. Warrant: theorem-grade per model as cited; the corridor is structural and its assumption is named.

The finite-verification wall

No instance mountain reaches a universal. The hypothesis is equivalent to a statement with a single universal quantifier over the integers and a decidable body, by the elementary reformulations of

Robin and Lagarias.^{16,17} No finite verification reaches such a statement. A universal over an infinite domain is not established by checking finitely many instances, and no premise about the instances being independent of one another is needed for that, nor would one be true here, since the divisor sum the body is built from is highly dependent across its argument. The field's own history is the proof that this is a wall and not a caution.

Box 3 | Three mountains that stood and fell

Each of the following ran clean on every value ever computed and was false past all computation, and two of the three imply the hypothesis. Polya's conjecture on the summatory Liouville function held at every computed point and fails first near nine hundred million, at 906,150,257.¹⁸ Mertens' conjecture sat far from its bound at every computed point and is false, its limit superior exceeding the bound beyond all computation.¹⁹ The sign of the prime-counting error, clean for as far as anyone has looked, reverses near a height on the order of ten to the three hundred and sixteenth.²⁰ Each was a mountain of verified instances that stood, and each fell. No mountain of instances reaches a truth about all integers, and the trillions of verified on-line zeros are a mountain of exactly that kind. They carry no evidential weight here, in either direction. Warrant: theorem-grade per item as cited; the inference to the wall is structural.

What the walls are and are not

The point the essay turns on is a single sentence stated at grade. That we cannot read the decree is a fact about our reach. It is not a fact that there is no fact. The veils bar our reading. They do not unwrite the thing read. The truth is determinate, one value, fixed by the primes, and it is unread. Unbounded is not undetermined. A universal over an unbounded domain has a single fixed answer whether or not any instrument reads it, and the wall that keeps us from checking it is a wall and not a room.

The third wall in particular bars exactly one thing and no more. It bars the infinite exhaustion of the cases. It does not bar a proof, which binds the whole tail in a single stroke the way induction settles a universal without visiting a large instance, and it does not bar a refutation, which needs one off-line zero exhibited and read. So from the wall it never follows that the decree is unprovable, only that no finite census will ever read it. The point is worth stating with force, because the opposite claim is more attractive than it deserves to be, and because the three things it might mean have three different answers. Unprovability comes in cells, and only one of them is safe.

The class-relative cell. The hypothesis may be unprovable in some particular theory, in Peano arithmetic or in set theory as it stands. This is the safe cell, it is exactly where the essay's verdict sits, and it decides nothing about the hypothesis, since a sentence unprovable in a given theory may be true or false.

The independence cell. The hypothesis may be independent of a sound theory, neither provable nor refutable there. This cell does not lean toward falsity. It leans the other way and it does so by proof. The negation of a statement of this logical form is witnessed at a finite

stage and would be provable in an extremely weak arithmetic the moment it held, so if the negation is not provable the statement is true. Independence forecloses falsity. To establish that a sound theory cannot refute the hypothesis is to establish the hypothesis one register up, which is why the independence cell is not a modest resting place either.

The unrestricted cell. One might assert that no proof can ever exist anywhere, in any sound system whatever. This is the cell where the claim collapses into its opposite, and the reason is the mirror of the reason above. If the statement is true then a proof of it exists in some sound system. The theory of the natural numbers is sound and proves every true sentence about them, this one included, and that alone carries the argument. A second illustration is often reached for here and it should be reached for with its caveat attached: the transfinite progressions of Turing and Feferman, obtained by iterating consistency or reflection along the ordinal notations, are complete for statements of this form.²⁴ The completeness is real and it is bought entirely by the choice of path through the notations, and no effective rule picks out a path guaranteed to reach only truths, so the theorem supplies no usable route to a proof for anyone who does not already know which path is right. The rung costs exactly the height gained. That caveat is why the progression is an illustration here and not a premise: delete the clause and the argument is untouched, since the theory of the natural numbers carries it alone. So to say that no sound system anywhere proves the statement is to say that it is not true. The unrestricted claim is the negation in costume, and it rides one floor, the determinacy of arithmetic truth, which is a premise this essay names and does not cross.

The three cells partition the ground and the essay stands in the first. It asserts unprovability in no theory, claims independence of none, and asserts the unrestricted form nowhere, because the second would decide the hypothesis true and the third would decide it false, and the essay decides neither. Unprovability in the unrestricted sense is not the safe agnostic position. It is a strong claim wearing modest clothes, and the modesty is what makes it worth naming.

Finally, the infinity at issue should be identified so that the staging is not mistaken for the content. The quantifier ranges over the integers, a countable unboundedness, and the continuum in which the strip is drawn is a stage on which a countable arithmetical fact is displayed. Strip the staging and the hypothesis is a determinate claim over the integers, a written thing behind a veil, and the veil is the reach of our instruments and not any softness in the object.

The obstructions, counted

Three walls is the count the essay began with and it is too coarse for what the classical record actually holds. The walls are three **non-entailments**, and they sit inside a wider structure of eleven distinct obstructions, each with its own mechanism, its own scope, and its own kind of permanence. Setting them out together is worth doing, because the differences between them are what a reader most often collapses, and because the last of them is of a different order from the other ten.

Each row names a classical fact or an elementary derivation, and the theological column is a reading laid beside it and never a step within it.
No row asserts the hypothesis and no row denies it.

Table 2A The eleven obstructions. Mechanism, scope, and the reading laid beside each.			
Obstruction	The mechanism	Its scope	Read theologically
The fold	a countermodel satisfies the fold axioms with zeros off the axis	the fold axioms alone	the decree is the axis, and a decree is not a census of who stands on it
The counting	a countermodel satisfies the Euler and counting axioms with zeros at the abscissa	those axioms alone	the outward regularity is not the inward one, and neither reads the other
The finite reach	no finite verification exhausts a universal over the integers	every finite instrument, always	the servant's reach is finite by constitution, and finitude is not a fault
The margin	zero-free methods yield a region and the claim asks for a line	every method of that family	approaching is not arriving, and no amount of approach becomes arrival
The proportion	lower bounds on a proportion do not reach a universal	every method of that family	a portion counted is not the whole beheld
The averaged sign	the departure is one-dimensional and comes in mirror families, so every invariant of the reflection averages the side away	every instrument built from the reflection	the veil is woven from the same symmetry that gives the decree, which is why lifting it is not a matter of looking harder
The equivalence	every published criterion restates the target rather than routing to it, and a restatement leaves no coordinate to gain on	all seven reformulations	a name repeated is not a thing given, and the tongue may hold a truth all night without conferring it
The missing base	the integers are initial in their setting, so their self-product collapses and the surface an index theorem would stand on is absent	that setting only, and the setting is not itself forced	a floor with nothing beneath it in one house, which is a fact about the house
The uniformity	the flow route needs a constant not degrading in its parameter, the coefficient route a bound uniform in degree, and neither is in hand	each route at its own frontier	the road is open and the distance is not yet crossed, which is a state of work and not of prohibition
The mirror	for a claim of this logical form, unrestricted unprovability is the negation and irrefutability by a sufficient system is a proof	any claim of the unrestricted form	the two despairing sentences are each a secret assertion, and neither is the humble one it dresses as
The grounding	no derivation grounds its own starting points, so every proof rests on posits put in place by an act	every proposition without exception, settled and open alike	the throne is empty by proof, and everything stands on it while it stands on nothing

Note: the margin and the proportion rows are two instances of one family, an approximation that improves toward a target and does not attain it, and they are listed separately because different repairs on different quantities would remove them. The first three rows are the non-entailments and the finite-reach wall, classical and cited below. The middle rows are elementary derivations. The mirror row is the mirror already stated. The final row is of a different order and is treated next. Warrant: structural as an arrangement, the individual results at their cited grades.

The last veil is not a veil over this hypothesis

The eleventh row differs from the other ten in a way that decides how it may be used, and the difference is worth stating plainly because it is the point at which a reading of this kind most easily overreaches.

The first eight are **reachability** obstructions. Each names an instrument, a method, or a route, and says that **this** does not reach **that**. Each is bounded, each has a scope, and one of them, the missing base, has an exit that has been attempted. They differ from one another and they differ in kind from the ninth.

The ninth is a **jurisdiction** obstruction and it is total. Nothing in mathematics is grounded in the sense it names, nothing ever has been, and the impossibility is a theorem rather than a gap in the record. It is the strongest obstruction in the list, it does not age, and no future instrument evades it, because it is phrased on the act of positing rather than on any method: whatever the coming instrument is, **using it will be a doing**, and a doing is not a derivation.

> And because it is total, **it singles out nothing**. A condition satisfied by no theorem separates no theorem from any other. The

ninth veil lies over the settled and the open alike, over arithmetic and geometry and set theory and over the function-field case that was proved, so it carries no information about which propositions are decided and which are not. **It is the strongest of the eleven and the only one that says nothing about this hypothesis in particular.**

The two facts hold together and neither weakens the other: the obstruction is absolute, and its absoluteness is exactly what makes it silent on particulars. Read as a veil over this hypothesis it would be a category error, mixing a bar on **level** with a bar on **reach**, and the essay declines it in that use while carrying it at full strength in its own.

Theologically the row reads as the emptiness of the throne, and the reading is offered as a reading. Everything stands on the ground and the ground stands on nothing, and that is not a defect in the ground but what being the ground means. The servant's proofs rest on posits the servant lays down, and laying them down is a deed. **That is the condition all of mathematics has always worked under, not a failure it has lately fallen into**, and the hypothesis is neither closer to nor further from a proof for it being so.

What cannot be summoned

Here the reading reaches its own centre, and the argument is about grounding rather than about witnessing. **The distinction matters, and stating it is what keeps the section standing.**

It is tempting to put the point in terms of a witness, and the temptation should be named and refused. A witness stands apart from what it witnesses, and one might say that nothing stands apart from the origin, so that only the origin could witness itself. **That argument does not hold, because it uses standing apart in two senses.** In the first it means role-separation, the verifier not being the claimant, and role-separation from the origin is ordinary: a reader examining the line is not the line. In the second it means being outside the origin's reach, which nothing is. **Neither sense makes both halves true**, and no witness in the history of mathematics has ever had the second kind of separation.

The point that does hold is about derivation, and it uses one sense throughout.

A derivation reaches its conclusion from premises it did not itself establish. That is what makes it a derivation rather than a stipulation, and it is why every proof stands on something laid down before it. **Now ask what premises would establish the origin.** They would have to come from somewhere the origin does not account for. **A second source would have to stand beside the first**, supplying what the first did not.

There is no second. So the premises that would ground the origin are not withheld, not hidden, and not merely undiscovered. **There is nowhere for them to come from**, which is the same sentence as the consequence stated earlier, said where a demand is being made rather than where a form is being placed.

And the claim is scoped exactly. It concerns grounding the origin, which nothing does. **It says nothing about whether anything**

stands off the line, which is a different question with a different answer, and one this reading leaves entirely where it found it.

The marginal line

The reading is interpretation, and it would be dishonest to dress it as more. But it is not unmoored, and there is a genuine structure in which the line behaves as an attracting configuration in the literal dynamical sense. Apply the heat flow to the completed zeta function, the one-parameter family obtained by evolving it under the backward heat equation. As the parameter increases the complex zeros migrate toward the line. There is a real number, the de Bruijn-Newman constant, marking the threshold: above it all zeros are on the line, below it some are not. The hypothesis is exactly the statement that this constant is at most zero. Newman conjectured the reverse inequality with the gloss that if the hypothesis is true it is only barely so, the zeros sitting exactly at the edge of alignment. Rodgers and Tao proved Newman's inequality in 2020.^{21,22} Combining the two, the hypothesis is equivalent to the constant being exactly zero, the zeros sitting precisely at the critical threshold of a flow that drives zeros toward the line.

This fits the reading with unusual precision. The flow is a genuine dynamics with the line as its attracting configuration, and the hypothesis says the actual function sits exactly at the marginal point where the driving has just succeeded, no earlier and no later. Alignment that is marginal, achieved exactly and with no slack, is the signature of a state reached by internal tendency rather than imposed by external force. A cage would hold the zeros with room to spare.

The typing of this section is stated so that it cannot be misread. It is container-grade. Delete it and no claim in this essay moves, which is the operational test of whether a passage is load-bearing, and this one carries no load. The flow concerns a deformation rather than the multiplicative structure directly, the equivalence leaves the hypothesis exactly as open as before, and the fit is suggestive and is not a proof. It is included because it shows the attractor language is not loose talk, and it is marked as removable because it is.

The barzakh

Barzakh is the partition, the isthmus set between two seas that meet and do not cross, the interspace that joins by keeping apart. The critical line is a partition in that exact sense. The functional equation divides the strip into two halves and the reflection carries each half onto the other. The line is the fixed set of that reflection, the one locus held invariant, the place the two sides meet and are not merged. It belongs to neither side and joins them by standing between.

Read as that fixed locus, the line has no thickness. It is a set of points, each its own image, and a locus of coincidence is not a measured band. Asked where the zeros must fall, the geometry answers with a coincidence, that each zero rest on its own reflection, and says nothing of a distance. And yet the line will not sit still under the eye. It has no thickness, and the whole field of values rests upon it. It stands half a unit from its mirror wall, a separation any ruler will read, and that half is itself a chart reading of the kind section four stripped away, the

number moving with the coordinates while the coincidence does not. Even so the reading is there to be taken, and the line has no breadth of its own to be the near side of the distance it is taken across. This doubleness is old, older than the hypothesis, the vertigo Zeno named, the whole made of parts that carry none of its measure. Where it lives is the thing to see. It belongs to the continuum, to the line rebuilt from points and handed a measure, present in the plainest interval exactly as it is here. Mathematics did not dissolve it. It settled it by a decision, letting length gather across intervals and refusing to let it gather across points, and a decision is a settlement and not a proof. The hypothesis borrows that ground to stand on, as every statement about the zeros must, and inherits the ground's strangeness the way a house inherits the tremor of the plain beneath it. The tremor is real and it is not a crack in the house.

The partition carries the veil, and here the barzakh reads doubly. A partition is a membrane a flow presses toward and stops at, which is the marginal structure of the previous section read in one word. But the barzakh is also the veil in the older sense, the interspace between the seen and the unseen, and read that way it is the exact figure of this essay. The truth of the hypothesis lies in the ghayb, the unseen, determinate and written and hidden. Our instruments press against the veil and do not cross it, the fold countermodel on one side, the counting countermodel in the middle, the finite wall on the other. The error refused in section twelve is the error of reading the barzakh as an open sea, a room to sail into forever, when it is a membrane over a written thing. The veil hides. It does not abolish what it hides.

The silent center

One feature of the formal situation has an old name in the theological register, and its scope has to be stated exactly, because the loose version claims far more than the theorem gives. Tarski proved that arithmetical truth is not definable by any arithmetical formula.²³ That is a statement about the global truth predicate and about nothing else. It says nothing whatever about any particular sentence, and the truth value of a single statement of the hypothesis's logical form is perfectly definable, by the statement itself if by nothing else. So Tarski forecloses the system naming truth as such, and Gödel forecloses completeness, and neither of them touches the provability status of this one sentence, which the walls above leave open and which this essay does not close.

What the theorem does supply here is a placement rather than a claim. The center the whole structure is organized around is present to the structure and silent to its global names: the ladder can climb toward truth and cannot assemble a name for truth out of its own symbols. That is the formal shadow of an old apophatic claim, that the ground is reached and not captured, approached and not pronounced. The shadow is worth seeing and it is worth exactly what it is, a statement about naming and not a statement about this sentence.

A grey zone lives here, and naming it precisely is the discipline of the section, because it is the easiest thing in the essay to misplace and the whole of the next section depends on placing it right. The grey zone is not in the primes. Every nontrivial zero has a definite offset, fixed by

the multiplicative structure before any reading begins, and the hypothesis has a definite truth value the primes already determine. What is grey is our access to that value from the structure we can see. The test that the greyness is in the reading and not in the object is decisive: it evaporates the instant the information arrives, a single computed off-line zero deciding the hypothesis at once, where a true indeterminacy in the object would survive any amount of learning. Indeterminacy that dies when one learns more was never in the thing. It was in the veil between what the structure shows and what is the case.

So the honest state of the hypothesis is a determinate truth we cannot read. Its truth value is fixed by the primes, one value, written. The witness that would let us read it is located, on the multiplicative axis neither channel reaches, and it is not in hand. Determinate, and unread. This state deserves its own name and is written here as terminal suspension, and it is not the ordinary open verdict. The ordinary open verdict says we do not yet know and may later, a gap the field expects to fill. Terminal suspension says the truth is fixed and written and the instrument has found the proven edge of its own reading, eleven counted obstructions and no twelfth thing to try along those roads, a sealed resolution boundary and not a gap of ignorance. The hypothesis is therefore not open in the ordinary sense. It is a written truth sealed away from our reading, and stating it exactly is what lets the next section refuse the two errors that misstate it in opposite directions.

The manufactured room and the premature seal

The determinate-and-unread state of the hypothesis invites two errors, and they are one error committed in two directions. The essay refuses both with equal force, and the refusal is symmetric, which is what keeps it honest rather than apologetic.

The first error. manufactures a room out of the veil. It reads our inability to read the decree as the absence of the decree. It treats the finite-verification wall, our inability to check every integer, as an ontological openness, a claim that there is no fact of the matter, and it treats the blindness of the two channels as a permission for the truth itself to be unsettled. Having read the veil as a void, it builds a room to dwell in, a perpetual openness, and takes up residence there. The structure of the error is exact and needs no attribution of motive to name. It is the category mistake of section three, mistaking a limit of the instrument for a property of the object, and it is the barzakh misread as an open sea. Unbounded is not undetermined. A truth we cannot finitely verify is still a truth with one value. The wall is a wall and not a room, and there is no room to inhabit, because there is only a written decree behind a veil we cannot cross.

The mildest version of the error is the one worth naming most carefully, because it is the version this essay's own earlier drafts committed. It takes a witness function that strays and reads its straying as a freedom the zeta zeros hold and decline. That reading is warm, and it is a leak. The witness is an instrument result. It says the channel is blind. To hear it as a permission held by the primes is to move a fact across a register boundary with nothing carrying it, and

the correction is not a weakening but a hardening: what the witnesses establish is non-compulsion, which is stronger and cleaner than an exercised alternative and is exactly what the decree requires. A decree that compels nothing needs no defector to prove it does not compel.

The second error: reads a located witness as a delivered proof, and it crosses the same veil from the other side. It points at the multiplicative axis where the deciding content lives and declares the decision made. It mistakes the witness being named for the witness being in hand, a located missing step for a taken step, and so announces a closure it has not earned. This is the triumphalist mirror of the manufactured room. Where the agnostic reads the veil as void, the triumphalist reads the location as a crossing, and both have confused a limit or a location for the thing itself. The essay refuses this exactly as it refuses the first. The witness is located and not supplied. To say where the decision lives is not to make it. No reading, this one included, crosses the veil, because crossing it is a proof, and this essay proves nothing.

The honest posture stands between the two and takes neither exit. It affirms the written decree and confesses the veil. The truth is determinate, one value, fixed by the primes. We cannot read it, by the fold countermodel, by the counting countermodel, and by the finite wall. We hold the verdict with no stake in its outcome, equally ready for either answer the moment the reading comes, a single computed off-line zero settling it against the line and a proof on the multiplicative axis settling it for the line, and we prefer neither in advance. This posture has a name in the tradition the essay reads from. It is tawakkul, the trust that the truth is fixed and held with Allah جل جلاله, and the refusal to manufacture from our own inability to read either a doubt or a vindication.

Here the symmetry of the theological quarantine does its work, and it is the answer to the whole worry that a mathematics could be leaned against a creed. The same discipline that forbids the agnostic to manufacture doubt from the veil forbids the believer to claim vindication from it. Theology does no mathematical work in this essay, and the doing-no-work runs in both directions. The wire is cut both ways. A reading that could pry the metaphysics open on the strength of the openness could equally close it on the strength of a proof, and this reading does neither, because the openness is a fact about our instruments and carries no metaphysical freight in either direction. That symmetric silence is not a hedge. It is the exact condition under which the reading is honest, and it is why the reading stands whatever the hypothesis turns out to be.

What the demand for a proof is a demand for

Here the formulation says what it is for.

A proof stands on an origin it does not derive. Its starting points are laid down and not obtained, and **no instrument reaches beneath the posits it is standing on**. Every instrument any of us holds stands on such posits already, which is why the reach is not a matter of effort.

To demand a proof of the decreed line is to demand an instrument that reaches beneath the origin and displays its

making. Not a better instrument. **A standpoint outside the origin**, from which the origin could be seen being laid.

And there is no outside. That is the whole content of the oneness, restated where the demand is made. **The formal domain cannot ground its own root. Syntax cannot verify the ground it stands on, the empty throne that everything rests upon and that rests on nothing**. These are not obstacles that a future construction removes. They are what it is to be inside a creation with a single source.

So the demand is not for more effort. It is for a place to stand that the oneness forecloses.

The ledger: one shape, eleven obstructions, one string, one door

The essay's findings can be set out in four rows, and the arrangement is the point. There is one settled shape, eleven counted obstructions of which three are the proven non-entailments and the last is of a different order, one determinate and unread proposition, and one located and uncrossed door. No row asserts the hypothesis and no row denies it.

The shape is settled. at theorem grade. The critical line is the fixed locus of the reflection composed from the functional equation and conjugation, elected before any zero and invariant under every chart, while the width of the strip, the address of the line, and the numerical value one half are readings taken off coordinates and move when the coordinates move. The composition is load-bearing and worth stating exactly, because the functional equation alone does not give the line. Its own fixed set is a single point: the condition that a coordinate equal one minus itself forces the value one half, and the condition that the other equal its own negative forces it to vanish. A point, not a line. The line appears only when the fold is composed with conjugation, which the completed function supplies through its reality on the axis, and the composite fixes the first coordinate at one half while leaving the second free. **The decree is the fold together with that reality, not the fold alone**. The half is its name in our chart.

The veils are settled. at theorem grade, each by an exhibited model rather than by an argument. The fold axioms have a model whose zeros leave the axis, the Davenport-Heilbronn function. The Euler-product-and-counting axioms have a model whose zeros run to the abscissa, the Diamond-Montgomery-Vorhauer system. And no finite verification reaches a universal over the integers, witnessed by three collapsed mountains. The first two together give the two non-entailments at theorem grade, each alone refuted as an entailment by one model, and neither refutation a near miss that a sharper version of the same argument could close. The corridor read off them, that a proof draws on both, is structural and carries the assumption that the two sets exhaust the resources, which nothing above supplies and which this essay's own door row declines. The third wall carries one qualifier and it is load-bearing: it is a wall against exhaustion of instances, never a demonstrated bar on proof, since a permanent

proof-bar on a statement of this logical form would itself very nearly decide the question.

The string is determinate and unread. Whether every nontrivial zero lies on the fold is one proposition with one truth value, fixed by the primes. It is not both true and false, and a proposition taken together with its negation selects nothing. It is terminal suspension in the sense fixed above, a written truth at a proven resolution boundary, and not the ordinary open question the field expects one day to fill by pushing along the same roads. The suspension is class-relative and stays there. No independence is claimed, since independence would decide the string true, and no unrestricted unprovability is claimed, since that would decide it false.

The door is located and uncrossed. Every road that would imply the hypothesis rather than restate it reduces to one requirement, deterministic control of the correlations of the primes with their own shifts. That this is one requirement in several disguises rather than several independent routes is what the published reformulations show when each is followed to where it binds.

Table 3 Seven costumes, one requirement. Where each reformulation binds.		
Reformulation	Where it binds	Reduces to
The analytic critical strip	its measure of distance is an artifact of the coordinates	the shifted correlations
Weil positivity	the explicit-formula positivity would need the primes to decouple, and they do not	the same correlations
The explicit formula	exact at each test function, carrying no uniform residue past it	the same correlations
Operator factorization	exhibiting the operator whose eigenvalues are the zeros is proving the object, not a route to it	the same correlations
The Nyman-Beurling distance	the Burnol floor: the distance is bounded below by the zeros themselves, so the criterion restates rather than routes	the same correlations
Symmetric reflection	the fold axioms have a countermodel, so no consequence of them alone can decide	the same correlations
The field with one element	the intersection-positivity that decides the analogue has no counterpart yet built over the integers	the same correlations

Note: Read the last column as one requirement in seven disguises. The table maps where the one road runs. It does not walk it. The contrast that gives the table its criterion is the one case on record that did lift: over a function field the deciding structure was a **general index theorem**, proved for its own reasons in its own setting and **not equivalent** to the statement it went on to prove, whose **specialisation** supplied the positivity. That is the shape a route must have. Every row above is a restatement of the target, and a restatement touches every coordinate the target has and therefore leaves none untouched to gain on. **A criterion equivalent to what it would decide is a translation, and translations do not lift.** Warrant: structural, the individual results theorem-grade as cited.^{25,26,27}

A witness that would supply the implication is constrained, and the constraints are classical and worth naming, because they say what a proof must look like and rule out what it cannot be. It must control the shifted correlations deterministically, not on average and not by any assumption of randomness, because the Diamond-Montgomery-Vorhauer system satisfies averaged and local conditions of the classical grade and still fails. It must be stable under forcing, because the hypothesis is a statement about all integers whose truth value forcing cannot alter, so the witness must live in arithmetic or arithmetic geometry and not in a set-theoretic extension. It must be genuine new arithmetic content and not a restatement of the fold axioms, which have a countermodel. And it must not smuggle multiplicative control over the additive correlations, asserting that unique factorization governs an off-diagonal it does not reach. A witness meeting these is a proof, which is why none is claimed here.

There is one place the analogous question is settled, and it shows where a witness would live. Over a curve over a finite field the analogue of the hypothesis is a theorem, the zeros appearing as eigenvalues of the Frobenius map on the cohomology of an actual geometric object and lying on the line because those eigenvalues are controlled at a single absolute value.^{28,29} The structure that decided that case is the multiplicative content made geometric, the correlations becoming intersection numbers on a surface whose positivity is a structural mandate rather than a combinatorial bound, and the reflection symmetry contributed nothing to the decision. The difficulty over the integers can then be named without discouragement. No geometric object over the integers is known to carry such a Frobenius, and building one, a geometry beneath the integers in which the correlations become intersections, is the explicit aim of the programs of Connes and Consani and their neighbors.³⁰ The deciding structure has been constructed once, geometrically, on the multiplicative axis, and the open problem is that the integers have not yet yielded their version of it. The impasse belongs to the two channels, not to the question.

Table 4 The ledger.			
Row	Content	Status	Warrant
The shape	the fold as fixed locus, elected before any zero, chart-invariant	settled	theorem
Wall one	the fold alone does not entail the conclusion, one countermodel, Davenport-Heilbronn	settled	theorem
Wall two	the Euler product plus classical counting does not entail it, one countermodel, Diamond-Montgomery-Vorhauer	settled	theorem
Two non-entailments	no consequence of the fold axioms alone decides, and none of the Euler-product-and-counting axioms alone, by two exhibited countermodels, closed under refinement	settled	theorem
The corridor	read off the two, that a proof draws on both, on the assumption that the two sets exhaust the resources	read, not established	structural
Wall three	no finite verification reaches the universal, three collapsed mountains	settled	theorem, with the no-proof-bar qualifier
The string	every nontrivial zero on the fold	determinate and unread	terminal suspension
The door	deterministic control of the shifted prime correlations	located, not supplied	structural

Note: No row asserts the hypothesis and no row denies it. The theological reading of the shape as qadar, of the nature as fitra, and of the resolution as tawhid is premise-grade throughout and load-bearing on nothing in any row.

Objections

The openness is a failure of grounding, and a properly grounded mathematics would close it. It would not, and the reason is that the grounding condition is satisfied by nothing at all. No derivation grounds its own starting points: a proof begins with axioms it cannot reach, and that no system establishes its own grounding from within is a theorem of others, Gödel's second incompleteness and Tarski's undefinability, not a position taken here²³. So every proof in the literature rests on posits it does not derive, and the posits are put in place by an act rather than obtained by one.

The consequence is exact and it cuts the objection off rather than answering it. **A condition satisfied by no theorem separates no theorem from any other.** Nothing in mathematics has ever been grounded in the sense the objection wants, so the want holds identically over the settled and the open, and it carries no information about which is which. It cannot be why this hypothesis is unproved, because it is equally true of every hypothesis that has been proved. Arithmetic rests on its axioms, geometry on its postulates, set theory on its own, the function-field case on the ambient geometry it was proved in, and none of them is self-grounding.

The honest statement is that mathematics has never operated under the stronger notion and so has not failed a standard it never adopted. A proof is complete relative to stated axioms, by definition, and every theorem there is has been produced under that notion. The reading

offered here works entirely inside it and claims nothing about the other.

The theology does mathematical work it has not earned. It does not, and the structure of the essay prevents it. Every mathematical assertion the reading uses is a classical theorem of others, cited below: the two countermodels and the non-entailments they establish, the localization to the Euler product, the reduction to a correlation positivity, the finite-verification wall and its collapsed conjectures, the on-line proportion, the heat-flow threshold. The theology adds no theorem and no estimate, and the hypothesis is exactly as open after the reading as before it. A description that does no mathematical work cannot smuggle a mathematical conclusion.

This is numerology. The number one half does no work in the reading, and section four shows why it cannot, since the value is an artifact of the chart while the fold is not. What the reading maps is a structure: a center elected before its occupants, a nature that is not compelled to it, and a conjecture that the occupants rest there of that nature. The mapping is onto a mathematical structure with the same logical shape, and it is offered as interpretation.

Teleology and decree have no place in mathematics. The reading does not claim the primes have purposes or that a will moves the zeros. The operative causation is the realization of an innate form, a structural relation requiring no intention, and the teleological vocabulary is eliminable in favor of the attractor structure of section nine and the formal-cause structure of section five. The vocabulary earns its place by making a structure visible and by naming why force-first methods fail.

If the hypothesis is false the reading collapses. Suppose it fails and some zero lies off the line. Then the multiplicative nature of the integers is not, after all, sufficient to bring the whole correlation to the fold, and the reading would say exactly that. This is not a collapse but

the determination of the reading's open conditional. The reading does not assert that the zeros are aligned. It asserts that the center is decreed, that nothing compels the occupants to it, that the truth about their occupancy is written and unread, and that if they rest there they rest there by nature. A false hypothesis answers the conditional in the negative and refutes no category.

The theological frame is parochial. Granted, and it is a clarification of status. The structure of a nature resolving to a decreed center can be read as a final cause in the Aristotelian register, as an immanent cause in the Spinozist one, as self-consistency in a purely formal one. These are translations, one structure through several windows, and the theological window is one faithful reading among possible faithful readings, the one the author writes from.

The lattice proves a proof must use both ingredients. It does not, and the essay declines the claim in its own sixth section. Two countermodels establish that each of two resource sets fails alone, which is theorem and is closed under refinement. They establish nothing about a resource reducible to neither, and this essay names a candidate for one in its thirteenth section, so the corridor is read off the lattice and is not entailed by it. To pass from each alone fails to both are required is to assume the toolbox is exhausted by the two tools in hand, and the assumption is supplied by nothing.

Refusing both escapes is agnosticism in disguise. It is not, and the distinction is exact. Agnosticism is a claim about the truth, that there is no fact or that the fact is unknowable in principle. The posture of section twelve is the opposite. It affirms that the truth is determinate, one value, fixed by the primes, and it is silent only about which value, and silent only because the witness is not in hand. It holds a determinate truth as unread, not an indeterminate truth as unsettled. And its silence is symmetric, forbidding manufactured doubt and premature vindication with one discipline, which is precisely what agnosticism does not do, since agnosticism takes doubt as its resting state.

The witnesses prove the zeros could have strayed. They do not, and this is the objection the essay is built to refuse. Each witness is a different function and neither shares the population whose behavior the hypothesis is about. What each establishes is that a set of axioms has a model in which the conclusion fails, which is a fact about the axioms. Converting that into a modal claim about the zeta zeros carries a fact across a register boundary with nothing to carry it. What the witnesses do establish, non-compulsion, is stronger and is all the reading requires.

The parity argument is doing the work and it is false. The parity argument is indeed false as a universal claim about functionals of the reflection, and section three states so and abandons it. The completed zeta function is sent to its own conjugate by the composed reflection, so its real part is even in the offset and its imaginary part is odd, and the imaginary part is a functional of exactly that data. The wall never rested on parity. It rests on a countermodel, which is a two-line argument and needs no claim about instruments at all. Parity

survives at its own address, governing the direction of a violation rather than its existence, and it is stated there and used nowhere else.

What the reading delivers and what it does not

It is essential to be exact, because the surrounding mathematics is the most consequential open problem in its field and the temptation to overstate is real.

Claimed, as theological interpretation of a settled structure.

A category for the hypothesis: a decreed center that compels nothing, met or not met by the innate nature of a population, read as qadar, fitra, and tawhid under one structure, the unity taken at three levels of which two are settled and the third is the open string, with the innate nature localized to the multiplicativity of the integers in particular and to the prime correlations where the decision lives. An explanation of the long failure of force-first programs as a category mistake, the search for an efficient cause where the operative cause is the innate form. The grounding structure: a derivation reaches its conclusion from premises it did not establish, and the premises that would ground a single origin would have to come from a second source that the oneness forecloses, which is a theological reading of the openness and not a claim about derivability. The witness framing that superficially resembles it is named and refused in the text, on the ground that it equivocates between role-separation and ontological externality. And a reading of the openness itself: the hypothesis is a determinate truth the primes hold and no finite instrument reads, terminal suspension rather than ordinary openness, veiled by eleven counted obstructions, the last of which is not about this object at all. The reading's centre is a single structure: a derivation reaches its conclusion from premises it did not establish, the premises that would ground a single origin would have to come from a second source, and there is no second, with two errors to be refused under one symmetric discipline and the honest posture being tawakkul. The lattice of section six is claimed at its split grade and no higher, the two non-entailments as theorem and the corridor read off them as structural with its one assumption named.

Not claimed. No proof of the hypothesis and no step toward one. No claim that the hypothesis is unprovable, which section eight shows to be a strong claim rather than a modest one, and which over function fields the settled analogue counts against. No claim that the zeros of the zeta function possess an alternative they decline, which no witness function establishes. No claim that a proof must draw on the fold and the Euler product, which two countermodels do not establish and which a third resource reducible to neither would falsify. No mathematics, the premises classical and cited. No mathematical work by theology, the reading removable in full without touching a proof, and the removal-without-effect symmetric, cutting equally against leaning the mathematics for the creed or against it. No settlement of the metaphysics.

The mathematical content of this essay is zero, by design and by honest accounting. Every mathematical assertion in it is a classical result of others, cited as such. What the essay contributes is a reading of those results, an account of the kind of statement the hypothesis is

and the kind of openness it carries. A reading is not a theorem. It can still be faithful or unfaithful, illuminating or idle, and the case made here is that this one is faithful, that it follows from what the witnesses actually establish rather than from what they are commonly heard to establish, and that it names, better than the force reading, both why a century of method has not closed the problem and why the openness is a veil and not a void.

A word on what sits beneath the reading. It rests on a structural claim about the formulation space rather than on any one formulation. The many published faces of the hypothesis are costumes on a single proposition. The obstructions are of two kinds, an instrument limit that relocates when the costume changes and a finite-verification limit that rides every costume by logical equivalence. And every road that would imply the hypothesis rather than restate it reduces to the one requirement Table 3 names. This is a map of where the roads run and where they stall, at structural grade, and it is itself no theorem. It decides nothing about the zeta function, it stands on classical results at every cited point, and it is open at its edges, foreclosing neither a formulation outside the survey nor a genuinely new road.

Conclusion

The Riemann Hypothesis has been approached for a century as a thing to be forced, a conclusion to be pressed out of the symmetry that fixes the critical line. The approach has not succeeded, and two counterexample functions show why it cannot. A function carrying the fold at one half and no Euler product places its zeros off the fold. A generalized prime system carrying an Euler product and no fold sends its zeros to the abscissa at one. Each is a model in which a set of axioms holds and the conclusion fails, so neither set entails the conclusion, which is theorem, and the corridor that reading opens, that a proof draws on both, is structural and carries its assumption openly, which is the split the essay's sixth section sets out as a lattice. Neither witness exhibits a freedom held by the primes. Each is a wall in front of the reader, and what the pair delivers is non-compulsion: the decree issues no order, and whatever holds the zeros to the line, if they are held, is the multiplicative nature of the integers themselves, reaching down to the correlations neither channel reaches.

Read through that structure, the line is the decreed center, qadar, the fold elected before any zero and by nothing in the plane it is drawn on, the one feature of the picture that survives every chart while the width and the address and the famous number do not. The hypothesis is the proposition that the primes keep faith with it, the whole correlation made one upon the one line, resting there not by force but by the innate nature of this population and not of its imitators. Qadar sets the measure. Fitra is the nature by which it is met, the multiplicativity of the integers, reaching each prime and, of itself, not the correlations where the meeting is decided. Tawhid is the many meeting it as one.

This is a reading and not a result. It settles nothing in the mathematics and adds nothing to it, and it draws its every mathematical premise from work that belongs to others. Its claim about the openness is the one it stands on. The hypothesis is a determinate truth the primes already hold, written and fixed, a terminal suspension and not the

ordinary open question, veiled three times: by a countermodel in the fold axioms that no sharper argument of the same kind can evade, by a second countermodel that holds even against multiplicativity and classical counting, and by a wall of finite reach the collapsed mountains prove is a wall and not a caution. That we cannot read the decree is a fact about our instruments and never a fact that the decree is unwritten. Unbounded is not undetermined.

Two errors are refused with one discipline. The manufactured room reads the veil as void and dwells in a perpetual openness that is not there. The premature seal reads a located witness as a delivered proof and announces a closure it has not earned. Between them stands the honest posture, tawakkul, the affirmation of the written decree and the confession of the veil, the verdict held with no stake and equally ready for either answer, the silence symmetric so that the mathematics leans neither for the creed nor against it. The decision lives on the multiplicative axis, where the analogue is already a theorem and the witness is located and not in hand. If the zeros come home to the line, they come home because the order of the primes is fine enough to bring them, each keeping faith with the center of its own accord, and the hypothesis is the conjecture that the order is that fine. Whether it is that fine is a fact about the primes, written and veiled, and its settlement rests with Allah جل جلاله.

Author's provenance and method disclosure

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full at the reference below. The paper rests solely on its cited results, every one of them the classical work of others, and the discipline that organized them contributes no theorem, no estimate, and no evidential weight in any direction.

Four derivations in the text are elementary and are carried out in place rather than cited, and they are named here so a reader can separate them from the classical results. The fixed set of the functional equation alone is computed and found to be a single point, so the axis is the fixed set of that map composed with conjugation. The direction of departure from the axis is computed and found to be one-dimensional, and the mirror families a strayed zero generates are exhibited. The claim that an equivalent criterion leaves no coordinate to gain on is argued from the definition of equivalence. And that no derivation grounds its own starting points is used as the theorem of others it is, cited at reference twenty-three, with only its consequence for discrimination drawn here. Each is checkable in a line or two and none requires the discipline that organized them.

References

1. Riemann, B. Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse. Monatsberichte der Berliner Akademie (1859).
2. Titchmarsh, E. C. and Heath-Brown, D. R. The Theory of the Riemann Zeta-Function, 2nd edn (Clarendon Press, Oxford, 1986).
3. Hamburger, H. Über die Riemannsche Funktionalgleichung der Zeta-Funktion. Math. Z. 10, 240-254 (1921).

4. Davenport, H. and Heilbronn, H. On the zeros of certain Dirichlet series, I and II. J. Lond. Math. Soc. 11, 181-185 and 307-312 (1936).
5. Kaczorowski, J. and Perelli, A. The Selberg class: a survey. In Number Theory in Progress (eds Györy, K. et al.) 953-992 (de Gruyter, Berlin, 1999).
6. Bombieri, E. and Ghosh, A. Around the Davenport-Heilbronn function. Russ. Math. Surv. 66, 221-270 (2011).
7. Voronin, S. M. On the zeros of some Dirichlet series lying on the critical line. Izv. Akad. Nauk SSSR Ser. Mat. 44, 63-91 (1980).
8. Balanzario, E. P. and Sánchez-Ortiz, J. Zeros of the Davenport-Heilbronn counterexample. Math. Comp. 76, 2045-2049 (2007).
9. Selberg, A. Old and new conjectures and results about a class of Dirichlet series. In Proc. Amalfi Conf. Analytic Number Theory 367-385 (Università di Salerno, 1992).
10. Kaczorowski, J. and Kulas, M. On the non-trivial zeros off the critical line for L-functions from the extended Selberg class. Monatsh. Math. 150, 217-232 (2007).
11. Diamond, H., Montgomery, H. and Vorhauer, U. Beurling primes with large oscillation. Math. Ann. 334, 1-36 (2006).
12. Hardy, G. H. Sur les zéros de la fonction $\zeta(s)$ de Riemann. C. R. Acad. Sci. 158, 1012-1014 (1914).
13. Conrey, J. B. More than two fifths of the zeros of the Riemann zeta function are on the critical line. J. Reine Angew. Math. 399, 1-26 (1989).
14. Montgomery, H. L. and Vaughan, R. C. Multiplicative Number Theory I: Classical Theory (Cambridge Univ. Press, 2007).
15. Aristotle. Physics II.3 and Metaphysics V.2. In The Complete Works of Aristotle (ed. Barnes, J.) (Princeton Univ. Press, 1984).
16. Robin, G. Grandes valeurs de la fonction somme des diviseurs et hypothèse de Riemann. J. Math. Pures Appl. 63, 187-213 (1984).
17. Lagarias, J. C. An elementary problem equivalent to the Riemann hypothesis. Am. Math. Mon. 109, 534-543 (2002).
18. Haselgrove, C. B. A disproof of a conjecture of Pólya. Mathematika 5, 141-145 (1958); Tanaka, M. A numerical investigation on the summatory function of the Möbius function. Tokyo J. Math. 3, 187-189 (1980).
19. Odlyzko, A. M. and te Riele, H. J. J. Disproof of the Mertens conjecture. J. Reine Angew. Math. 357, 138-160 (1985).
20. Skewes, S. On the difference $\pi(x) - \text{li}(x)$, II. Proc. Lond. Math. Soc. 5, 48-70 (1955); Bays, C. and Hudson, R. H. A new bound for the smallest x with $\pi(x) > \text{li}(x)$. Math. Comp. 69, 1285-1296 (2000).
21. de Bruijn, N. G. The roots of trigonometric integrals. Duke Math. J. 17, 197-226 (1950); Newman, C. M. Fourier transforms with only real zeros. Proc. Am. Math. Soc. 61, 245-251 (1976).
22. Rodgers, B. and Tao, T. The de Bruijn-Newman constant is non-negative. Forum Math. Pi 8, e6 (2020).
23. Tarski, A. Der Wahrheitsbegriff in den formalisierten Sprachen. Stud. Philos. 1, 261-405 (1936); Gödel, K. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. Monatsh. Math. Phys. 38, 173-198 (1931).
24. Turing, A. M. Systems of logic based on ordinals. Proc. Lond. Math. Soc. 45, 161-228 (1939); Feferman, S. Transfinite recursive progressions of axiomatic theories. J. Symb. Log. 27, 259-316 (1962).
25. Nyman, B. On the one-dimensional translation group and semigroup in certain function spaces. Thesis, Uppsala (1950); Beurling, A. A closure problem related to the Riemann zeta-function. Proc. Natl Acad. Sci. USA 41, 312-314 (1955).
26. Burnol, J.-F. A lower bound in an approximation problem involving the zeros of the Riemann zeta function. Adv. Math. 170, 56-70 (2002).
27. Li, X.-J. The positivity of a sequence of numbers and the Riemann Hypothesis. J. Number Theory 65, 325-333 (1997); Báez-Duarte, L. A strengthening of the Nyman-Beurling criterion for the Riemann hypothesis. Rend. Lincei Mat. Appl. 14, 5-11 (2003).
28. Weil, A. Sur les courbes algébriques et les variétés qui s'en déduisent (Hermann, Paris, 1948).
29. Deligne, P. La conjecture de Weil I. Publ. Math. IHÉS 43, 273-307 (1974).
30. Connes, A. and Consani, C. Geometry of the arithmetic site. Adv. Math. 291, 274-329 (2016).
31. The Qur'an. Verses referenced for the concepts invoked: 2:256 on the absence of compulsion, 25:53 and 55:19-20 on the barzakh between the two seas, 30:30 on the fitra as the innate disposition, 33:72 on the trust borne by the free, and the verses on the ghayb and on tawakkul.
32. Islam, M. TRISDUCTION: A Linguistically, Topologically, and Mathematically Sealed Verification Architecture. Triaxial Orthogonality, Twelve-Gate Closure, the Quaternionic Completion, the Root Axiom, and the Master Pre-Sealed Proposition Ledger. Zenodo, version 4, 19 June 2026. DOI 10.5281/zenodo.20757507. <https://zenodo.org/records/20757507>. Mirror: PhilArchive record ISLTTG, <https://philpapers.org/rec/ISLTTG>. Master reference, continuously updated at the same location.
1. Hadamard, J. Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques. Bull. Soc. Math. France 24, 199-220 (1896); de la Vallée Poussin, C. J. Recherches analytiques sur la théorie des nombres premiers. Ann. Soc. Sci. Bruxelles 20, 183-256 (1896).
2. Levinson, N. More than one third of zeros of Riemann's zeta-function are on $\sigma = 1/2$. Adv. Math. 13, 383-436 (1974).
3. Csordas, G., Norfolk, T. S. & Varga, R. S. The Riemann hypothesis and the Turán inequalities. Trans. Amer. Math. Soc. 296, 521-541 (1986).
4. Griffin, M., Ono, K., Rolen, L. & Zagier, D. Jensen polynomials for the Riemann zeta function and other sequences. Proc. Natl. Acad. Sci. USA 116, 11103-11110 (2019).

Appendix T. A Formal Proof of Riemann Termination, with a Cascade Specification of Termination

Abstract. The Riemann Hypothesis is terminally unreachable by the formal-alone instrument **as swept**; the unrestricted form of that sentence is priced at Section 12.3 and is not asserted here. We establish an equivalence governing its formal status: **universal unprovability of the Riemann Hypothesis over the class $\mathcal{T}$ of consistent recursively axiomatized extensions of Robinson arithmetic $\mathbf{Q}$ is materially identical to its strict refutation in $\mathbf{Q}$.** That equivalence is purely proof-theoretic and uses no soundness hypothesis. Under Σ_1^0 -soundness of $\mathbf{Q}$ it sharpens: universal unprovability then equates to falsity in the standard model $\mathbf{N}$. Only the sharpening uses soundness. **The even functionals of the frame are blind to the sign** by Proposition 5.2; λ is not, by Proposition 5.3. We specify a runnable eight-gate cascade to adjudicate the termination of the instrument's search space, on a four-dimensional real inner-product space split as $E_+ \oplus E_-$ by a self-adjoint involution. **Its condition count is exactly eight, and once two stipulations are made the count is a theorem and not a further stipulation.** The stipulations are that the decomposition carries three atomic axes and that an atom admits negation and no partial operation. Given them, the group carrying each axis of the residence to itself is $(\mathbb{Z}/2)^3$, of order eight, with no ninth element; the argument runs through stabilisation of the decomposition and **not** through invariance of the Gram determinant, which is invariant under a much larger group and is disowned as a criterion. **No record satisfying all eight conditions is exhibited here**, Ξ_2 and Ξ_4 being uninstantiated in fact, and no completed sweep is claimed. What is claimed is the boundary: an emission may be terminal **for the search space as swept** and may not be terminal unrestrictedly, since by the main equivalence any clause entailing universal unprovability entails $\mathbf{Q} \vdash \neg R$. The frame carries **one polarity bit**, invisible to the even functionals and recovered by λ , **determined by the syntax of $\mathbf{P}$** under Proposition 6.1a and computed by the instrument; it bears on ζ not at all.

1. Introduction

Two questions about the Riemann Hypothesis are routinely conflated. The first is whether R is provable. The second is whether an instrument, having exhausted a catalog of approaches, may declare that exhaustion permanent.

Section 3 settles the first. Theorem 3.2 shows that *no formal system proves R* is not a modal statement at some remove from arithmetic but is equivalent, over $\mathcal{T}$, to Q refutes R ; and Corollary 3.6a sharpens this, under Σ_1^0 -soundness of Q , to equivalence with the falsity of R . Sections 6 through 8 settle the second: we specify formally when an exhaustion verdict is admissible, prove the specification fail-safe and cardinality-bounded under stated hypotheses, and prove that the permanence clause it requires cannot be strengthened to a universal-unprovability claim without becoming a refutation.

The mathematics of Section 3 is elementary and classical in its ingredients. What is offered is the equivalence in the form given, the sharpened Π_1^0 form, and the specification of Sections 6 through 8 in which the consequence becomes a stated admissibility condition.

2. Preliminaries

2.1 Arithmetic. Q denotes Robinson arithmetic. $\mathcal{T}$ is the class of theories T with T consistent, T r.e., and $T \supseteq Q$. *Consistency is meant and soundness is not.*

2.2 Σ_1^0 -completeness (classical). Q proves every true Σ_1^0 sentence.

2.3 Σ_1^0 -soundness. Q is Σ_1^0 -sound: every Σ_1^0 sentence Q proves is true in $\mathbb{N}$. This holds because $\mathbb{N} \models Q$. Where this is used it is named.

2.4 The Riemann Hypothesis in Π_1^0 form. *Lagarias (2002):* R holds iff for every $n \geq 1$, $\sigma(n) \leq H_n + \exp(H_n) \ln(H_n)$, with $H_n = \sum_{k \leq n} 1/k$. *Robin (1984):* R holds iff for every $n > 5040$, $\sigma(n) < e^{\gamma} n \ln \ln n$.

The classification runs on the Lagarias form only. Write $R = \forall n \varphi(n)$ with $\varphi(n)$ the Lagarias inequality $\sigma(n) \leq H_n + \exp(H_n) \ln(H_n)$. **We do not claim φ is decidable:** the comparison is against a transcendental-valued expression and deciding a non-strict inequality requires excluding exact equality, which is not available for $n \geq 2$. What holds, and suffices throughout, is that **φ is Π_1^0** : its negation $\sigma(n) > x$ is semi-decidable by evaluating x at increasing rational precision, so φ is Π_1^0 and by quantifier contraction R is Π_1^0 with $\neg R \Sigma_1^0$.

2.4a The sentence is fixed, and nothing transfers across formalizations. From here on **R denotes one fixed sentence**, namely

$$R := \forall n \forall k [\sigma(n) \leq q(n,k) + 2^{-k}],$$

where **$q(n,k)$ is any fixed primitive recursive rational-valued function satisfying the approximation guarantee**

$$|q(n,k) - (H_n + \exp(H_n) \ln(H_n))| \leq 2^{-k} \text{ for all } n \geq 1 \text{ and } k \geq 1,$$

which one such procedure meets by evaluating H_n exactly, computing $\exp$ and $\ln$ at $k + 40$ places, and truncating to a multiple of 2^{-k} ; executed, the ratio $|q - r|/2^{-k}$ reaches **0.9534** and never 1 over $n \in \{1, 12, 60, 5040, 55440\}$ and $k \in \{1, 5, 10, 20, 40\}$. **The guarantee is what makes the displayed sentence well defined**, and without it the sentence names nothing in particular.

Lemma 2.4b (the displayed sentence is the Lagarias criterion). Write $r_n = H_n + \exp(H_n) \ln(H_n)$. Then $\sigma(n) \leq r_n$ iff $\forall k [\sigma(n) \leq q(n,k) + 2^{-k}]$.

Proof: $(\implies) r_n \leq q(n,k) + 2^{-k}$ by the guarantee, so $\sigma(n) \leq r_n$ gives $\sigma(n) \leq q(n,k) + 2^{-k}$ for every k . $(\impliedby) q(n,k) \leq r_n + 2^{-k}$ by the guarantee, so the hypothesis gives $\sigma(n) \leq r_n + 2 \cdot 2^{-k}$ for every k , and letting $k \rightarrow \infty$ gives $\sigma(n) \leq r_n$. ■

Hence R holds iff the Lagarias inequality holds at every n , which by Lagarias (2002) holds iff the Riemann Hypothesis does. **The matrix is decidable:** it compares an integer with an explicitly computed rational, so R is Π_1^0 and $\neg R$ is Σ_1^0 by quantifier contraction, genuinely and not merely as a limit of approximations, and Q 's Σ_1^0 -completeness applies to $\neg R$ as stated at 2.2. Executed: the displayed form agrees with the direct comparison $\sigma(n) \leq r$ at $n = 1, 12, 60$ and 5040.

Two consequences bind and are stated because they are easy to lose. First, **every theorem of Sections 3, 4 and 7 is about this sentence** and about no other. Second, **the equivalence between R and the Riemann Hypothesis is Lagarias's theorem of ordinary mathematics and is not a theorem of Q** . Two Π_1^0 sentences true-equivalent over $\mathbb{N}$ need not be equivalent over a weak base, so $Q \vdash \neg R$ is a claim about this arithmetization, and it reaches the statement that ζ has a zero off the critical line **only through Lagarias's equivalence, in the metatheory, and never inside Q** . Wherever this paper prices an unprovability claim as that statement, including at 12.3, the pricing carries that step and it is named here once rather than assumed throughout.

The semi-decidability step, stated in full, since it is the one step a reader is likely to reconstruct. The right-hand side is a **computable real**: there is a computable sequence of rationals q_k with $|q_k - r| < 2^{-k}$, obtained by evaluating H_n exactly and $\exp$ and $\ln$ to $k + 1$ places. For rational x ,

$$x > r \text{ iff } \exists k (x > q_k + 2^{-k}),$$

which is Σ_1^0 . $(\Leftarrow) r < q_k + 2^{-k} < x. (\Rightarrow) \text{ put } \delta = x - r > 0 \text{ and choose } k \text{ with } 2^{-k} < \delta/2; \text{ then } q_k < r + 2^{-k} < x - \delta/2, \text{ so } q_k + 2^{-k} < x$. Dually $x \leq r$ iff $\forall k (x \leq q_k + 2^{-k})$, which is Π_1^0 . Since $\sigma(n)$ is an integer and H_n rational, $\varphi(n)$ is Π_1^0 and $\neg\varphi(n)$ is Σ_1^0 , and **no decidability of φ is claimed or needed**: the equality case is exactly where no finite precision settles the comparison.

Executed illustration of the search that Σ_1^0 names. On the true instances the negation is false and the search correctly does not halt: at $n = 1, 12$ and 5040 the margins are 0, 0.3218372596 and 492.3187310894 and no k satisfies the condition. On a synthetic excess above the bound it halts at a computable k : excess 0.5 at $n = 12$ halts at $k = 2$, excess 10^{-2} at $k = 7$, and excess 10^{-6} at $n = 5040$ at $k = 20$. **No actual violation is exhibited and none is known**, since exhibiting one would refute R ; the halting behaviour is therefore shown on a synthetic excess, which is what makes the negation Σ_1^0 rather than what makes it true.

The Robin form is cited for context and is **not** used for the classification. Its inequality is strict, so $\sigma(n) < x$ is Σ_1^0 and its negation is not, placing the Robin matrix in Σ_1^0 and the criterion in Π_1^0 . One Π_1^0 form is all that is required.

2.5 Executed computation of the matrix at finite precision. Computed at 80-digit decimal:

| n | $\sigma(n)$ | $H_n + e^{H_n} \ln(H_n)$ | margin | |---|---|---| | 1 | 1 | 1.00000000000000000000000000000000 | 0 | | 12 | 28 | 28.32183725964540620598233 |
0.32183725964540620598 | | 60 | 168 | 170.9766842646928136888768 | 2.97668426469281368887 | | 5040 | 19344 | 19836.31873108944797731162 |
492.318731089447977311 | | 55440 | 232128 | 241179.9211023638387578728 | 9051.92110236383875787 |

Robin form at 40 digits: margin 14128.08735 at $n = 5041$, 565.5185643 at $n = 10080$, 90021.39744 at $n = 720720$. These are finite-precision evaluations, not decisions of ϕ , and they bear on R not at all. They are given because a Σ_1^0 certificate of a violation, if one existed, would take this form.

2.6 Linear algebra. For an arbitrary real 3×3 matrix M and $D = \text{diag}(\epsilon_1, \epsilon_2, \epsilon_3)$ with $\epsilon_i \in \{\pm 1\}$, D is orthogonal with $D = D^T = D^{-1}$ and $\det(D) = \epsilon_1 \epsilon_2 \epsilon_3$.

3. The Universal Unprovability Theorem

Definition 3.1. For a sentence R of the language of Q put $U(R) := \forall T \in \mathcal{T} (T \not\vdash R)$.

Theorem 3.2 (Main). For every sentence R , $U(R) \iff Q \vdash \neg R$.

Proof. ($\implies$) Assume $U(R)$. If $Q + R$ were consistent then, being r.e. and extending Q , it lies in $\mathcal{T}$ and proves R , contradicting $U(R)$. So $Q + R$ is inconsistent and $Q \vdash \neg R$ by the deduction theorem.

($\impliedby$) Assume $Q \vdash \neg R$ and let $T \in \mathcal{T}$. Then $T \vdash \neg R$ since $T \supseteq Q$; if also $T \vdash R$ then T is inconsistent, contradicting $T \in \mathcal{T}$. Hence $T \not\vdash R$. ■

Remark 3.3. Theorem 3.2 is an equivalence and asserts neither of its sides. Its proof uses no soundness assumption, no semantic notion, and no property of R beyond its being a sentence.

Remark 3.2a (what drives the equivalence, said plainly). The forward direction turns on a single membership: **$Q + R$ is itself one of the theories quantified over.** It is finitely axiomatized, hence r.e., extends Q , and proves R , so if it is consistent then $U(R)$ fails. That is the whole mechanism. The equivalence is therefore a formal consequence of admitting $Q + R$ among the candidates, it is elementary, and **it supplies no independent mathematical evidence bearing on R .** What it does supply is a price: an unprovability claim quantified over that class is not a weaker statement than a refutation, it is the same statement.

Corollary 3.4. $U(R)$ implies $Q + R$ is inconsistent.

Corollary 3.5 (Π_1^0 case). For $R = \forall n \phi(n) \in \Pi_1^0$: $U(R) \iff Q \vdash \exists n \neg \phi(n)$. Since Q is Σ_1^0 -sound (2.3), $U(R)$ further implies that some n_0 with $\neg \phi(n_0)$ exists; $\neg \phi$ being Σ_1^0 , the certificate for n_0 is verifiable by a terminating computation. **The cost of the claim is therefore concrete and should be read as such:** asserting universal unprovability of R over $\mathcal{T}$ has, under 2.3, exactly the arithmetic content of asserting that a specific pair (n_0, k_0) exists with $\sigma(n_0) > q(n_0, k_0) + 2^{-k_0}$, together with the obligation to exhibit it. It is not a weaker or more cautious claim than a counterexample; it is a counterexample claim with the witness left unstated.

Corollary 3.6 (Riemann, syntactic form). For R the Riemann Hypothesis, $U(R) \iff Q \vdash \exists n (\sigma(n) > H_n + \exp(H_n) \ln(H_n))$.

Corollary 3.6a (Riemann, sharpened form). Using Σ_1^0 -soundness of Q (2.3): for every Π_1^0 sentence R ,

$U(R) \iff \Omega \models \neg R$.

In particular $U(R)$ holds **for this R** if and only if $\mathbb{N} \models \neg R$; and $\mathbb{N} \models \neg R$ holds if and only if ζ has a zero off the critical line, **that last step being Lagarias's theorem in the metatheory and not a theorem of Q** per 2.4a.

Proof. By Theorem 3.2, $U(R) \iff Q \vdash \neg R$. If $Q \vdash \neg R$ then by Σ_1^0 -soundness $\mathbb{N} \models \neg R$. If $\mathbb{N} \models \neg R$ then $\neg R$ is a true Σ_1^0 sentence, so $Q \vdash \neg R$ by Σ_1^0 -completeness. ■

Remark 3.6b. Corollary 3.6a is the sharpest form of the paper's thesis: *no consistent r.e. extension of Q proves R if and only if R is false*. Note the hypothesis: Theorem 3.2 is soundness-free, and 3.6a is not.

Theorem 3.7 (Dichotomy). Let $R \in \Pi_1^0$, let $\mathcal{S} \subseteq \mathcal{T}$ be nonempty, and **suppose $Q + R$ is consistent**. Put $U_{\mathcal{S}} := \forall T \in \mathcal{S} (T \not\vdash R)$. Then exactly one of:

(a) **$Q + R \in \mathcal{S}$.** Then $U_{\mathcal{S}} \iff Q \vdash \neg R$. (*The implication is vacuous, and its vacuity is its content: $Q + R \in \mathcal{S} \subseteq \mathcal{T}$ gives $Q + R$ consistent, hence $Q \not\vdash \neg R$, while $Q + R \vdash R$ gives $\neg U_{\mathcal{S}}$ outright. **The horn cannot be occupied at all**, which is the sharper reading of Remark 3.8's claim that asserting $U_{\mathcal{S}}$ here converts the claim into $Q \vdash \neg R$: the conversion is available and the conjunction is not.*)

(b) **$Q + R \notin \mathcal{S}$.** Then $\mathcal{S}$ is fixed by a condition beyond consistency, r.e.-ness and extension of Q , and establishing $T \not\vdash R$ for any $T \in \mathcal{S}$ yields $\mathbb{N} \models R$ by Theorem 4.1.

If $Q + R$ is inconsistent then $Q \vdash \neg R$ outright and no dichotomy arises.

Proof. (b) is the negation of (a) under the hypothesis, so exactly one obtains. In (a) the argument of Theorem 3.2($\implies$) applies with $\mathcal{S}$ for $\mathcal{T}$. In (b), $Q + R$ is consistent, r.e. and extends Q , so its exclusion is by a further condition; the second clause is Theorem 4.1, and every $T \in \mathcal{S} \subseteq \mathcal{T}$ satisfies $T \supseteq Q$. ■

Remark 3.8. A reader will observe that no one asserting the unprovability of R intends to quantify over theories that assume R . The disjunction *either $Q \vdash \neg R$ or $\mathbb{N} \models R$* holds for every Π_1^0 sentence R by Σ_1^0 -completeness alone (2.2), independently of Theorem 3.7. **What Theorem 3.7 adds is that neither horn of a restricted unprovability claim can be occupied without paying one side of that disjunction:** horn (a) converts the claim into $Q \vdash \neg R$, horn (b) converts its companion half into $\mathbb{N} \models R$.

4. The Restriction Theorem

Theorem 4.1. Let $R \in \Pi_1^0$ and let $T \supseteq Q$ be any theory. If $T \not\vdash \neg R$ then $\mathbb{N} \models R$.

Proof. Write $R = \forall n \phi(n)$ with $\phi \in \Pi_1^0$. If $\mathbb{N} \not\models R$ then $\neg \phi(n_0)$ for some n_0 ; $\neg \phi(n_0)$ is Σ_1^0 , so by Σ_1^0 -completeness of Q we get $Q \vdash \neg \phi(n_0)$, hence $Q \vdash \neg R$, hence $T \vdash \neg R$. Contrapositively, $T \not\vdash \neg R$ implies $\mathbb{N} \models R$. ■

Remark 4.1a. No hypothesis on T beyond $T \supseteq Q$ is used: T need not be consistent, r.e., sound, or Σ_1^0 -complete.

Corollary 4.2. If $T \not\vdash R$ and $T \not\vdash \neg R$ then $\mathbb{N} \models R$.

Corollary 4.3. For any $T \supseteq Q$, in particular PA or ZFC or ZFC with large cardinals, a proof that T does not refute R establishes R .

Remark 4.3a (the two results are compatible, and here is why). A reader may put Theorem 3.2 beside Theorem 4.1 and see a tension: the first says universal unprovability yields $Q \vdash \neg R$, the second says any $T \supseteq Q$ failing to refute R yields $\mathbb{N} \models R$. **These cannot conflict, because their hypotheses are exclusive.** For $R \in \Pi^0_1$, exactly one of two cases holds. If $Q \vdash \neg R$ then every $T \in \mathcal{T}$ refutes R , so no T fails to refute it and Theorem 4.1's hypothesis is never met; under 2.3, R is false. If $Q \not\vdash \neg R$ then $Q + R$ is consistent, r.e. and extends Q , so it lies in $\mathcal{T}$ and proves R , so $U(R)$ fails and Theorem 3.2's antecedent is never met; and by Theorem 4.1, R is true. **Universal unprovability and non-refutation by some extension cannot both hold**, and nothing in this paper asserts that they can.

Remark 4.4 (the pincer). Unrestricted, the claim that the route to R is closed is equivalent to $Q \vdash \neg R$ by Theorem 3.2 and, under 2.3, to $\mathbb{N} \models \neg R$ by Corollary 3.6a. Restricted to a fixed $T \supseteq Q$, establishing its refutation-side half yields $\mathbb{N} \models R$ by Theorem 4.1. No formulation both quantifies non-trivially and leaves the truth-value of R untouched.

5. The Sign Character of the Frame Functional

5.1 What this section is for. No condition of Section 6 tests λ . Ξ_7 compares two recorded matrices entrywise; λ appears in Section 6 only inside Ξ_7 's consequence note, where it is defined and used to state a consequence of the condition and is not itself a clause of it. What Section 5 supports is the polarity account at Section 12.3 and the consequence bound at Ξ_7 's note; it is a supporting section and is stated as such.

Notation. For $Q_0 \in \mathbb{R}^{3 \times N}$ with unit rows, write $\mathbf{R_G} := Q_0 Q_0^T$ for the Gram matrix, reserving R for the Riemann Hypothesis. Let E be an orthonormal basis of the row span and put $\lambda(Q_0) := \det(Q_0 E^T)$.

Proposition 5.2. For arbitrary real 3×3 M and $D = \text{diag}(\epsilon_1, \epsilon_2, \epsilon_3)$, $\epsilon_i \in \{\pm 1\}$: $\det(D M D) = \det(M)$.

Proof. $\det(D M D) = \det(D)^2 \det(M) = \det(M)$, since $\det(D) \in \{\pm 1\}$. ■ (No symmetry or positive-semidefiniteness is used.)

Proposition 5.3 (sign character). $\lambda(DQ_0) = \det(D) \lambda(Q_0)$, and hence $|\lambda(DQ_0)| = |\lambda(Q_0)|$.

Proof. $\lambda(DQ_0) = \det(DQ_0 E^T) = \det(D) \det(Q_0 E^T) = \det(D) \lambda(Q_0)$. ■

Corollary 5.4. The Gram matrix of DQ_0 is $D \mathbf{R_G} D$, with $\det(D \mathbf{R_G} D) = \det(\mathbf{R_G})$ by 5.2; and for the total-negation pattern $D_- = -I_3$ one has $(-Q_0)(-Q_0)^T = \mathbf{R_G}$ exactly, while $\lambda(-Q_0) = -\lambda(Q_0)$.

5.5 Executed check. Over 200 unit-row triads at $N = 24$ and all eight sign patterns: $\max|\det(D \mathbf{R_G} D) - \det(\mathbf{R_G})| = 0.000 \times 10^0$, and $\max|\lambda(DQ_0) - \det(D) \lambda(Q_0)| = 0.000 \times 10^0$. Over 20000 triads, $\max|\lambda^2 - \det(\mathbf{R_G})| = 4.330 \times 10^{-15}$.

6. The Cascade: Formal Specification

Remark 6.0a (what this section does not do, stated before it is done). Four readings of Section 6 are natural, are wrong, and are foreclosed here rather than downstream, because each has been arrived at independently by a careful reader.

No text is mapped into $\mathbf{V}$, and no proposition is rotated. The atoms of (L1) index the axes of 6.1(i); they are not vectors and nothing is added, scaled, or rotated in the language. What occupies the space is the recorded matrix Q_0 of 6.1(viii), whose row index is the atom index and whose entries are numbers. No embedding of syntax into a vector space is offered, none is used, and none is needed: the axes are labelled slots and the algebra runs on the recorded frame, not on the sentence.

The conditions test the record and not the object. Every Ξ_i except Ξ_7 reads fields the record's author supplied, and Remark 6.6a states this at its least favourable. **The emitter performs no measurement of the Riemann Hypothesis and none is claimed.** What it certifies is that a sweep was declared, executed and reported to a stated discipline; that is the whole of its office, and Theorem 6.9 is the statement of exactly how little that entails.

The frame's orientation is independent of the truth value of P , by construction, and that is the design and not a defect. Proposition 6.1a builds Q_0 from the syntax of P alone, so the polarity bit it carries is the parity of leading negations and nothing else. A reader who observes that the geometry never touches the arithmetic has read the construction correctly; Theorem 6.9(a) is that observation proved, and Section 12.3 records that nothing about ζ is at stake in the bit.

What the geometry is for is answered at Remark 6.8a, and the answer is narrow: it supplies an exhaustiveness argument for the count of conditions and nothing further.

A record can be supplied for any proposition, including a trivial one, and would emit. Nothing prevents it and nothing should: the terminal token reports that *this instrument swept this* declared catalog and found the door uncrossed. It is not a difficulty rating, not a certificate about the object, and not evidence that the proposition resisted anything. A token on a trivial sentence would be a true report about a trivial sweep.

Definition 6.0 (the linguistic layer, stated in place). Definition 6.1(i) and 6.7a through 6.8 presuppose a decomposition discipline on the object propositions. It is stated here rather than cited, so that every hypothesis of Theorem 6.7 is checkable within this paper. Nothing outside those passages uses it. Four clauses.

(L1) Atomic decomposition. A complete factual proposition P is split into three components: A_1 its existence component, what is; A_2 its kinetic component, what it does or undergoes; A_3 its relational component, how it stands toward what.

(L2) Deletion test. Delete each component in turn. A complete atomic claim returns exactly **three** irreducible slots, each leaving an incoherent residue when removed. A count other than three is a failure of the decomposition at this layer.

(L3) Isolation. The vocabularies populating the three slots are **pairwise disjoint**. If one slot's vocabulary is a subset of another's, the slots are not independent and the decomposition fails.

(L4) Atomic bivalence. An atom admits negation and admits no partial or continuous operation. Correspondingly the verdict economy of Definition 6.3 admits no fractional or probabilistic state.

(Clauses (L1) through (L4) are stipulations of this construction, operational and reproducible across analysts applying them, and they are typed at that level: **they are not offered as a uniqueness theorem of first-order logic and no such theorem is claimed.** Every use of them below is marked. Theorem 6.7(ii)'s operational derivation of $d = 3$ uses (L2); its algebraic derivation uses none of them.)

Definition 6.1 (record). Fix a language $\mathcal{L}_{\text{obj}}$ for the object propositions and a disjoint language $\mathcal{L}_{\text{rec}}$ for records; **no field of a record is a sentence of $\mathcal{L}_{\text{obj}}$ or encodes one**, except as expressly noted in the witness fields, which are governed by Ξ_5 and by 2.4's classification.

An *audit record* r for a proposition $P \in \mathcal{L}_{\text{obj}}$ consists of:

- (i) a **labelled orthogonal decomposition** into atomic axes, $\mathbb{R}_{e_1} \oplus \dots \oplus \mathbb{R}_{e_d}$, one axis per atom of (L1), with d recorded; the space $V := \mathbb{R} \cdot u \oplus (\mathbb{R}_{e_1} \oplus \dots \oplus \mathbb{R}_{e_d})$ for a unit u orthogonal to every e_i ; and the involution σ **constructed from that decomposition**, $\sigma(u) = u$ and $\sigma(e_i) = -e_i$ for each i . (σ is not recorded as a primitive. The decomposition is the recorded object and σ is read off it, so $E-(\sigma)$ is the span of the atomic axes by construction and $\dim \text{Fix}(\sigma) = 1$ and $\sigma^2 = I$ hold by construction. Ξ_3 below therefore verifies that the construction was carried out and that $d = 3$; it does not test independent facts about an independently supplied involution. The row index of the frame Q_0 at (viii) is this same atom index.)
- (ii) a quantifier classification of P in the arithmetic hierarchy with its citation;
- (iii) a **declared register** ρ , its acting group $\mathbb{G}_\rho$, the invariant ring of $\mathbb{G}_\rho$, and the finite token list of the claim;
- (iv) a finite catalog $\mathcal{C}$ of *channels*, $|\mathcal{C}| = C$, each carrying a designated reduction with **its citation set**, together with a **closure flag** for $\mathcal{C}$;
- (v) a subset $\mathcal{M} \subseteq \mathcal{C}$, $|\mathcal{M}| = B$, each element carrying a **named obstruction and its warrant grade**;
- (vi) a finite set $\mathcal{P}$ of *roads*, **defined: each channel of (iv) determines exactly one road, namely the citation set recorded for its reduction; distinct channels determine the same road when their citation sets coincide; hence $|\mathcal{P}| \leq C$** . Two roads are *premise-disjoint* when they share no citation. Ξ_6 's floor of three therefore strengthens Ξ_3 's rather than restating it, and since $|\mathcal{P}| \geq 3$ with $|\mathcal{P}| \leq C$ gives $C \geq 3$, Ξ_3 's **channel floor $C \geq 3$ is entailed by Ξ_6 and is retained as a guard that halts earlier in the evaluation order of Definition 6.3**. It is therefore not independent of Ξ_6 , which is recorded here and at 9.9 rather than left for a reader to derive; the count of eight is a count of conditions evaluated, not a claim of pairwise independence;
- (vii) **witness fields** for P and $\neg P$, each either empty or holding a certificate; a **declared search budget** $k = (k_n, k_{\text{prec}}, k_{\text{pf}})$, **each component a standard numeral**, bounding the numeral range, the rational precision and the derivation length independently, with $k_n \geq k_{\text{min}}$, $k_{\text{prec}} \geq p_{\text{min}}$, $k_{\text{pf}} \geq d_{\text{min}}$ for declared floors recorded on the record; a **declared theory $\mathfrak{B}$ with $\mathfrak{B} \supseteq Q$** in which the proof search runs; **the floors are free parameters and are declared as such at 9.9, so the certificate clause of Ξ_3 is informative only to the extent the floors are, and a record declaring minimal floors emits a correspondingly weak statement; a completion flag** for each search; and an independence field;
- (viii) a **frame assignment** $Q_0(\cdot)$ recording a unit-row matrix $Q_0(P) \in \mathbb{R}^{\{d \times N\}}$ with its row-span dimension recorded, and, independently and with no constraint imposed here, a matrix $Q_0(\neg P) \in \mathbb{R}^{\{d \times N\}}$; a declared tolerance τ , and a **declared consequence ceiling β with $\beta \leq \beta_{\text{max}}$** . (No basis, Gram or orientation quantity is a field: none is consumed by any condition, and $\lambda(P)$ is undefined on a record declaring row-span dimension 2. G and λ are defined inline at Ξ_7 's consequence note, the only place they are used.)
- (ix) a designated *door*, an object not in $\mathcal{C}$, with a **provenance flag** valued *supplied* or *generated* and a **crossed flag**;
- (x) a permanence clause.

Definition 6.2 (the eight conditions).

- Ξ_1 the recorded axis count d equals 3, hence $\dim V = 1 + d = 4$. ($\sigma^2 = I$ and $\dim \text{Fix}(\sigma) = 1$ are invariants of 6.1(i)'s construction and **not tests**: they hold at every d , executed at $d = 2, 3, 4, 5$, so a condition asserting them would be analytically true on every record and would discriminate nothing, which is the defect Remark 6.2a records for a superseded form of Ξ_3 and Remark 6.1a-rank for a superseded clause of it. **$\det(\sigma|E-)$ is not among them**: σ acts on $E-$ as $-I_d$, so $\det(\sigma|E-) = (-1)^d$, which is -1 exactly when d is odd, executed at $d = 2, 3, 4, 5$ returning $+1, -1, +1, -1$. It is therefore a parity test and not an invariant, and it does not pick out $d = 3$ in any case, since it passes at every odd d . Ξ_3 's single clause $d = 3$ entails it at the value used and subsumes it. The division-algebra structure noted at Remark 6.7e is neither a record field nor tested by any condition.)
- Ξ_2 the quantifier class of P is located by a cited classical equivalence, and every token of (iii) lies in the invariant ring of $\mathbb{G}_\rho$.
- Ξ_3 $C \in \mathbb{N}$, $C \geq 3$, every channel in $\mathcal{C}$ carries a reduction to a common invariant by a cited classical equivalence with none unreduced, and the closure flag of (iv) is set. **Closure here means the catalog is displayed and fully reduced, not that no channel exists outside it**; the two readings are distinguished at Corollary 8.2 and the weaker one is meant, which is what makes Ξ_3 jointly satisfiable with Ξ_8 's door.
- Ξ_4 $B = C$, and each element of $\mathcal{M}$ carries its named obstruction with grade recorded.
- Ξ_5 the certificate search for $\neg P$ over numerals $n \leq k_n$ at rational precision k_{prec} returned no certificate; the proof search for P in the declared theory $\mathfrak{B}$ over derivations of length $\leq k_{\text{pf}}$ returned none; $k_n \geq k_{\text{min}}$, $k_{\text{prec}} \geq p_{\text{min}}$ and $k_{\text{pf}} \geq d_{\text{min}}$ with all six recorded; both completion flags are set; both witness fields are empty; and the independence field is empty.
- Ξ_6 $|\mathcal{P}| \geq 3$, and the roads in $\mathcal{P}$ are pairwise disjoint **as computed from the recorded citation sets**, with the recorded flag agreeing. (The clauses $B/C = 1$ and $C \geq 3$ are entailed by Ξ_4 and Ξ_3 and are not restated here.)
- Ξ_7 the recorded frames satisfy the negation convention **as tested directly**: $\max|Q_0(\neg P) + Q_0(P)| \leq \tau$ for a declared tolerance τ recorded on the record with $2\sqrt{N} \cdot \tau + N \cdot \tau^2 \leq \beta$ and $\beta \leq \beta_{\text{max}}$ for a declared ceiling β_{max} , and **the recorded row-span dimension of $Q_0(P)$ equals d** . (β bounds the consequence deviation directly and supersedes a separate tolerance ceiling: bounding τ alone left τ and N free to be declared large

together.) (The ceiling is required for the same reason the budget floors are: an unbounded τ admits vacuous satisfaction. $\beta_{\max}$ joins the census in its place.) (Consequence, stated as a bound rather than as an equality. Corollary 5.4 gives $G(\neg P) = G(P)$ and $\det(Q_0(\neg P)E(P)^T) = -\lambda(P)$ **at $\tau = 0$ and there only**. For $\tau > 0$, writing $Q_0(\neg P) = -Q_0(P) + \Delta$ with $\max|\Delta| \leq \tau$ and $Q_0(P)$ unit-row, one has $\max|G(\neg P) - G(P)| \leq 2\sqrt{N}\tau + N\tau^2$, and the orientation ratio deviates from -1 by $O(\sqrt{N}\tau|\lambda(P)|)$. Executed at $N = 24$: deviation 3.603×10^{-8} at $\tau = 10^{-8}$ and 3.342×10^{-2} at $\tau = 10^{-2}$, and at $N = 240$ with $\tau = 10^{-2}$, 4.385×10^{-2} ; at $\tau = 0$, exactly 0.000×10^0 . **The β clause bounds this deviation directly, which is why no separate tolerance ceiling is carried.** The rows of $Q_0(\neg P)$ are within $\sqrt{N}\tau$ of unit, not within τ : an entrywise perturbation of size τ moves a row norm by up to $\|\Delta\|_2 \leq \sqrt{N}\tau$. Executed: max row-norm deviation 2.0148×10^{-2} at $N = 24$ and 2.2149×10^{-2} at $N = 240$, both at $\tau = 10^{-2}$, against $\tau = 10^{-2}$ and $\sqrt{N}\tau$ of 4.899×10^{-2} and 1.549×10^{-1} . Nothing consumes the row norms. Here and only here, $G(X) := Q_0(X)Q_0(X)^T$ and $\lambda(X) := \det(Q_0(X)E^T)$ for E an orthonormal basis of the row span of $Q_0(P)$. **E must be the same on both sides:** for any such E one has $\lambda(-Q_0) = (-1)^3\lambda(Q_0) = -\lambda(Q_0)$, so the identity is basis-independent when shared and fails when recomputed per side, the identity of E being a gauge. Nothing in this paper depends on which E is chosen.)

- Ξ_8 the door of (ix) lies outside $\mathcal{C}$, its provenance flag reads *supplied*, its crossed flag is unset, and the permanence clause is present.

Proposition 6.1a (a frame assignment satisfying Ξ_7 exists). A hash applied directly to the syntax of P will not do: the hash of " $\neg P$ " is uncorrelated with the hash of P , so $Q_0(\neg P) = -Q_0(P)$ would have to be imposed by hand rather than obtained, and double negation would be inconsistent. Executed, the direct construction violates the convention on **6 of 6** test propositions. The polarity normal form obtains it instead. Put P into that form: let P_+ be the negation-free core and $s \in \{0,1\}$ the parity of leading negations. Define $Q_0(P) := (-1)^s \cdot H(P_+)$, where H seeds a deterministic generator with a hash of the syntax of P_+ and returns a row-centred, row-normalized $3 \times N$ draw. Then Q_0 is a total function of syntax, independent of the truth value of P , satisfies $Q_0(\neg P) = -Q_0(P)$ for every P including under double negation, and yields a record satisfying Ξ_7 .

The conclusion holds for any draw of rank d , which the construction achieves with probability one; a draw of lower rank records row-span dimension below d , fails Ξ_7 's dimension clause, and is rejected and reseeded.

Verification against the condition as it now stands, six propositions at $N = 24$ with their negations and double negations: $\max|Q_0(\neg P) + Q_0(P)| = 0.000 \times 10^0$, which satisfies Ξ_7 for every $\tau \geq 0$; recorded row-span dimension 3 in every case; and $\max|Q_0(P) - Q_0(\neg\neg P)| = 0.000 \times 10^0$. ■

Remark 6.1a-rank. 6.1(viii) records the row-span dimension of $Q_0(P)$; Ξ_7 tests whether it is d . E is not a field and is introduced at Ξ_7 's consequence note, where alone it is used. (Verified: rank 3 and $\lambda(P) \neq 0$ agree over 2000 frames with zero disagreements, and a forced rank-2 frame gives $|\lambda(P)| = 0.000 \times 10^0$.)

Remark 6.1b. Proposition 6.1a exhibits one assignment satisfying Ξ_7 and shows that condition satisfiable. It does not claim the assignment canonical, and no result depends on which **syntax-determined** assignment is used.

It does not show $\Xi_1, \Lambda \dots \Lambda \Xi_8$ satisfiable, and this paper does not. No record is exhibited. **This is a vacuity question about the specification and is not a hypothesis of any theorem:** Theorem 6.9(a) is P -invariance of satisfaction for the seven conditions other than Ξ_8 , which holds for every record whether or not any record satisfies all eight. What an unsatisfiable conjunction would cost is that no emission ever occurs, so 6.9(b) and (c) would speak of nothing.

What would discharge it, and this is a list of what is not done rather than a partial discharge. $\Xi_1, \Xi_3, \Xi_5, \Xi_6, \Xi_7$ and Ξ_8 are instantiable from material this paper carries: a four-dimensional space with a self-adjoint involution of one-dimensional fixed space; three channels with cited classical equivalences; a budget of the size exhibited at 2.5, with its floors; roads as defined at 6.1(vi); the assignment of 6.1a; and a door with clause W . **Two are not.** Ξ_2 requires an exhibited register with its acting group, which a trivial group would satisfy degenerately and which is not exhibited here. Ξ_4 requires a named obstruction with a recorded warrant grade for each channel; **this paper names no obstruction and will not invent one**, which is why the record is not constructed rather than being constructed badly.

Remark 6.2a (what Ξ_7 is, and why it is a condition). Ξ_7 compares the two independently recorded frames directly at a declared tolerance, so a record whose $Q_0(\neg P)$ differs from $-Q_0(P)$ by more than τ fails it. **Ξ_7 tests the frame assignment and not the proposition**, and no information about P is obtained from it. Testing the derived pair instead, equal Gram with opposite orientation, would be strictly weaker: that pair is implied by the convention and does not exhaust it, and two counterexample families exist, $Q_0(\neg P) := Q_0(P)(I_N - 2vv^T)$ for any unit v in the row span, and a single-axis reflection in the eigenbasis of $G(P)$.

Definition 6.3 (the emitter). (By Remark 6.0a the conditions below read record fields, Ξ_5 excepted; the emitter measures the declared sweep and not the object. Ξ_3 's floor $C \geq 3$ is entailed by Ξ_6 with $|\mathcal{A}| \leq C$ and is evaluated here **as a guard that halts earlier**, not as an independent condition; see 9.9.) Evaluate $\Xi_1, \Xi_2, \Xi_3, \Xi_4, \Xi_5, \Xi_6, \Xi_7, \Xi_8$ in order. If Ξ_5 fails because a witness field is populated, exit to direction-adjudication; if because the independence field is populated, exit to the relative-independence verdict. Otherwise any failure halts with the non-terminal verdict and the index recorded. If all hold, return the terminal token with the warrant cap and the permanence clause of Ξ_8 .

Proposition 6.4 (fail-safe, general form). If any field of Definition 6.1 required by some Ξ_i is absent, the emitter returns the non-terminal verdict.

Proof. By inspection over the eight conditions. $\Xi_1, \Xi_2, \Xi_3, \Xi_4, \Xi_6, \Xi_7, \Xi_8$ each require an equality, a set inequality, or a flag to be set; an absent field satisfies none. Ξ_5 requires k_{n,k_prec}, k_{p_f} and their three floors recorded with each bound at or above its floor, and both completion flags set; absence of any of these fails Ξ_5 , rather than passing it. The three floors are free parameters and are declared as such in 9.9. ■

Remark 6.4a. Ξ_5 requires an executed search reported against recorded bounds rather than an unpopulated field, so absence of data fails it, and the fail-safe property holds at every condition.

Proposition 6.5. The terminal token is returned only if all eight conditions hold.

Proof. Immediate from sequential evaluation and halt on first failure. ■

(Proposition 6.5 asserts the necessity direction only. Logical independence of the conditions is not proved here; see 9.9.)

Proposition 6.6. Ξ_5 is the unique condition that can route to a verdict other than the non-terminal one. **It can also fail to the non-terminal verdict**, when a bound or floor is unrecorded, a completion flag unset, or a bound below its floor; the exit branches are taken only when a witness field or the independence field is populated.

Proof. By inspection of Definition 6.3. ■

Remark 6.6a (which conditions reach outside the record, and it is fewer than one would like). The honest split is four ways. Ξ_5 alone reaches into arithmetic: its certificate clause reports an executed search over numerals, and Theorem 6.9(b) names it the section's one object-language output. Ξ_2 and Ξ_3 reach outward only through their citation requirements, a located quantifier class and cited classical equivalences, which a reader can check against the literature. Ξ_6 is computable from recorded data and is the one condition where a declaration can be caught wrong by the emitter: disjointness is computed from the citation sets and checked against the recorded flag. **The remaining four are internal consistency checks on declared fields:** Ξ_1 checks the recorded d and the construction invariants of 6.1(i), Ξ_4 reads a recorded obstruction and grade, Ξ_7 compares two recorded frames, and Ξ_8 reads recorded provenance and crossed flags.

A flag set by the record's author is not external material, however the relation it names is described. This remark exists to state the asymmetry, so it states the least favourable reading the specification supports.

Definition 6.7a (the decomposition and its stabiliser). A decomposed proposition is not a bare frame in V . By (L1) and (L3) it is **three labelled orthogonal axes**, one per atomic component, with pairwise disjoint vocabularies. By 6.1(i) the residence is that decomposition: $E_-(\sigma) = \mathbb{R}e_1 \oplus \dots \oplus \mathbb{R}e_d$, one axis per atom of (L1), with the frame's row index the same atom index. Define its **stabiliser**

$$\text{Stab}(E_-) := \{ g \in O(E_-) : g(\mathbb{R}e_i) = \mathbb{R}e_i \text{ for each } i \},$$

the transformations carrying each axis to itself, hence carrying a decomposed proposition to a decomposed proposition.

Lemma 6.7b (the stabiliser is exactly the sign changes). $\text{Stab}(E_-) = \{ \text{diag}(\epsilon_1, \dots, \epsilon_d) : \text{each } \epsilon_i \in \{+1, -1\} \} \cong (\mathbb{Z}/2)^d$, of order 2^d .

Proof. If $g(\mathbb{R}e_i) = \mathbb{R}e_i$ then $ge_i = \lambda_i e_i$ for some real λ_i ; $g \in O(E_-)$ forces $|\lambda_i| = 1$, hence $\lambda_i = \pm 1$, hence g is diagonal with entries ± 1 . Conversely every such diagonal matrix is orthogonal and fixes each axis setwise. ■

Remark 6.7b' (why stabilisation and not invariance of the determinant). The Gram determinant is invariant under a far larger group: $\det((gQ_0)(gQ_0)^T) = \det(g)^2 \cdot \det(Q_0 Q_0^T)$, so every g with $|\det g| = 1$ leaves it fixed, an eight-dimensional group containing $O(3)$ and the signed permutations of order $2^3 \cdot 3! = 48$. Executed against a reference frame at $\det(R_-G) = 0.931277980144$: the eight sign changes deviate by 0.000×10^0 , the six row permutations by 2.220×10^{-16} with unit rows preserved to 1.1×10^{-16} , five hundred random elements of $O(3)$ by at most 2.442×10^{-15} .

Determinant-invariance is therefore not the criterion and is not used as one. The criterion is Definition 6.7a, and the two families above fail it for reasons this paper states elsewhere. A rotation by $\pi/7$ in the e_1 – e_2 plane carries e_1 out of $\mathbb{R}e_1$, mixing one atomic slot into another and destroying the pairwise vocabulary disjointness (L3) requires; the image is not a decomposed proposition. A row transposition carries $\mathbb{R}e_1$ to $\mathbb{R}e_2$, so it does not fix each axis either, and by (L3) it is not a mere relabelling: it substitutes one atom's content for another's across disjoint vocabularies, which is content-mixing on exactly the ground the rotation is excluded. **Both are excluded by Definition 6.7a on its own terms and no appeal to gauge is made or needed.**

Definition 6.7c (complete screen). A family of conditions is a **complete screen** for the decomposition when it carries exactly one condition per element of $\text{Stab}(E_-)$. (Each element is one pattern of negation across the atoms. The identity is the empty pattern, P itself, and screens no conflation; of the 2^d elements, $2^d - 1$ move the frame. The count is of patterns, not of conflations.)

Remark 6.7c' (why the eight do not collapse to four). The enumeration at Theorem 6.7(iv) below reads $1 + 3 + 3 + 1$, which is also the orbit decomposition of $(\mathbb{Z}/2)^3$ under the permutation action of S_3 on the axes. The arithmetic is correct and the collapse is not available. **S_3 does not act on the screened objects.** By (L3) the three axes carry pairwise disjoint vocabularies, so the three weight-one patterns deny three different atoms and are three different propositions: denying what stands, denying what moves, and denying how they bind are not one object seen three ways. The weight grading is a way of displaying the eight, not a quotient of them.

Theorem 6.7 (given d axes the count is 2^d , and at $d = 3$ the eighth is the last). Let $E_-(\sigma) = \mathbb{R}e_1 \oplus \dots \oplus \mathbb{R}e_d$ be the labelled orthogonal decomposition of 6.1(i). Then:

(i) $|\text{Stab}(E_-)| = 2^d$, by Lemma 6.7b;

(ii) hence a **complete screen carries exactly 2^d conditions**, and at $d = 3$ exactly **eight**;

(iii) the eight enumerate by negation weight as **$1 + 3 + 3 + 1$** : the empty pattern, three denials of a single atom, three of a pair, and one total negation. **The eighth is the last**, and the ladder terminates because it has exhausted the atoms;

(iv) **no ninth element of $\text{Stab}(E_-)$ exists at $d = 3$** . A ninth requires $2^d \geq 9$, hence $d \geq 4$, hence a fourth atomic axis.

Proof. (i) is Lemma 6.7b. (ii) from (i) and Definition 6.7c. (iii) is the expansion of 2^3 by Hamming weight, with no weight 4 available at $d = 3$. (iv) $2^3 = 8 < 9$. ■

Remark 6.7d (what the count counts). Ξ_7 compares two recorded frames and Ξ_6 computes disjointness from the recorded citation sets; neither carries information about P , and by Theorem 6.9(a) the only condition that does is Ξ_5 , whose content is the search report. Theorem 6.7 fixes **how many** conditions a complete screen carries and says nothing about the discriminating power of any one of them over P .

Remark 6.7e (what is stipulated and what is proved). **Two things are stipulated.** $d = 3$, carried by (L2); and atomic bivalence, carried by (L4), which is what motivates restricting the axis-fixing group to $O(E_-)$ and so makes it finite: the deletion test returns three irreducible slots on a complete atomic claim, at the operational grade of Definition 6.0 and not as a uniqueness theorem. **The identification of the atoms with the axes of the residence is definitional**, by 6.1(i), where σ is constructed from the decomposition rather than recorded independently; there is no bridge to supply and none is claimed. Given those, **the count is not a further stipulation**: it is Lemma 6.7b, which is theorem-grade, together with the arithmetic of (iv).

The Frobenius classification is **corroboration and is load-bearing on nothing**. At $d = 3$ the space V of 6.1(i) has dimension 4, and $\mathbb{H}$ is the unique real associative division algebra with more than one imaginary unit; but no definition, condition or proof in this paper uses a multiplication on V , no record field carries a division-algebra structure, and Ξ_1 does not test one. The observation is recorded because it is true and because a reader will notice the dimension, not because anything rests on it.

Remark 6.8 (the shape of the claim). The claim this section makes is narrower than a forcing and wider than a choice, and it is stated at its exact strength. **Two things are stipulated:** that the decomposition carries three axes, which is (L2); and that atoms admit negation and no partial operation, which is (L4), motivating Definition 6.7a's restriction to $O(E_-)$, which is what makes the group finite. Both are stipulations of Definition 6.0 at the grade stated there. **One thing is definitional:** that the residence is the span of the atomic axes, by 6.1(i). **Everything after that is theorem.** By Lemma 6.7b the group carrying each axis to itself is exactly $(\mathbb{Z}/2)^d$; by Theorem 6.7 a complete screen has 2^d members, at $d = 3$ exactly eight, and no ninth. **Once the stipulations are made the count is not a further stipulation, and that is the whole of what is claimed here.**

No appeal to gauge is made. Both excluded families fail Definition 6.7a directly, and by Remark 6.7c' the permutation action does not reach the screened objects, so the eight do not collapse to four.

What remains open. Which condition screens which element: $\text{Stab}(E_-)$ acts on itself simply transitively, so no element is distinguished by the group structure alone, any bijection serves equally, and nothing below depends on which is taken. And the choice to screen the axis-fixing group rather than some larger structure-preserving group is a decision of this construction, motivated by (L3) and (L4) and not derived from them. **This paper does not claim that determinant-invariance forces eight**, and by Remark 6.7b' it does not: that group is eight-dimensional.

Remark 6.8a (why a space and not a checklist, and what the apparatus costs). A reader who has followed Remark 6.0a will ask what the geometry is for, since the eight conditions read record fields and could be evaluated by a boolean checklist with no space, no involution and no frame. The question is fair and the answer is narrow.

The geometry buys exactly one thing: an argument that the count is complete. A checklist of eight items carries no reason that it is eight rather than five or thirty, and no reason that a ninth is unnecessary rather than merely unwritten; its cardinality is whatever its author stopped at. Lemma 6.7b and Theorem 6.7 supply what a checklist cannot: given d axes, the group carrying each to itself is exactly $(\mathbb{Z}/2)^d$, so a complete screen has 2^d members, at $d = 3$ exactly eight, and no ninth element of that group exists. That is an exhaustiveness argument for the list, and it is available only because the conditions are indexed by a group rather than enumerated by hand.

It buys nothing else, and this should be weighed rather than glossed. The space performs no measurement of the object, per Remark 6.0a. Ξ_7 is the only condition that touches the frame, and Ξ_7 exists because the frame does. Strip the geometry and seven of the eight conditions survive unchanged as record checks, the eighth disappears with the frame it compares, and what is lost is the completeness argument and only that. **Whether one exhaustiveness argument is worth a four-dimensional apparatus is a judgement this paper leaves to the reader, and it states the trade rather than obscuring it.** The alternative reading, that the specification is an administrative protocol with a group-theoretic warrant for the length of its checklist, is not a misreading and is not resisted here.

Theorem 6.9 (payload). Let r be an audit record for a proposition P . Then:

(a) **whether r satisfies $\Xi_1, \Xi_2, \Xi_3, \Xi_4, \Xi_6, \Xi_7$ and Ξ_8 is independent of the truth value of P ;** and the satisfaction of Ξ_5 depends on it in exactly one way, namely through whether the executed certificate search returned a certificate at the declared budget. Consequently $\Xi_1 \wedge \dots \wedge \Xi_8$ entails no $\mathcal{L}_{\text{obj}}$ -sentence undecided in $\mathfrak{B}$, and hence neither does any emission on a record satisfying them;

and if in addition r satisfies Ξ_1 through Ξ_8 and the emitter returns the terminal token with permanence clause W , then:

(b) the emission entails that $\mathcal{C}$ is closed and fully reduced with $B = C$, that a door outside $\mathcal{C}$ is supplied and uncrossed, and, **the one object-language clause the emission makes explicit, that the executed certificate search returned no certificate at numerals $n \leq k_n$ and precision k_{prec} .** (It does not entail that no certificate exists there: that would require the search to be sound and complete for its budget, and the record carries no such field. Theorem 8.1 applies to the emitter here, since a search is an instrument at the register.) This clause is a finite verified computation, hence $\mathfrak{B}$ -provable, so (a) is unaffected, the clause being decided in $\mathfrak{B}$;

(c) if W is replaced by a clause entailing $U(P)$, the emission entails $Q \vdash \neg P$ by Theorem 3.2.

Proof. (a) Take r fixed and vary the truth value of P . $\Xi_1, \Xi_2, \Xi_3, \Xi_4, \Xi_6$ and Ξ_8 are conditions on fields of types (i)–(vi), (ix)–(x), none of which is or encodes an $\mathcal{L}_{\text{obj}}$ -sentence, so their satisfaction is untouched. Ξ_7 compares $Q_0(\neg P)$ to $Q_0(P)$ and to the recorded dimension, mentioning P nowhere beyond its index. That is the seven.

Ξ_5 is not among them and its dependence is stated rather than argued away. Its certificate clause requires that the executed search *returned* nothing, and what a search returns is a fact about the executed computation, which depends on whether a violating numeral lies at $n \leq k_n$ and precision k_{prec} . Three assignments make this concrete: P true, the search returns nothing and Ξ_5 passes; P false with the witness beyond the budget, the search returns nothing and Ξ_5 passes; **P false with the witness inside the budget, the search returns a certificate and Ξ_5 fails.** Satisfaction of Ξ_5 is therefore *compatible* with either truth value and is not *independent* of it.

The stated conclusion follows without closing that gap. Ξ_5 's content is *the search returned nothing at (k_n, k_{prec})* , a terminating computation and hence decided in $\mathfrak{B}$ either way; the other seven contribute nothing about P by the paragraph above. So the emission entails no $\mathcal{L}_{\text{obj}}$ -sentence that $\mathfrak{B}$ leaves undecided, which is (a) as stated.

(The stronger two-model claim, that $\Xi_1 \wedge \dots \wedge \Xi_8$ entails neither P nor $\neg P$, would require a satisfying record under each truth value, which is at least as strong as satisfiability and is disclosed as open at 6.1b. The restriction above is what the proof gives and it carries no undischarged hypothesis.)

(b) The first two clauses from Ξ_3, Ξ_4 and Ξ_6 ; the third from the certificate clause of Ξ_5 read as a report of the executed search, whose content is a terminating computation and therefore $\mathfrak{B}$ -provable. (c) By Theorem 3.2 applied to P . ■

7. The Termination Theorem

Definition 7.1. A *permanence clause* delimits the scope of an emitted token's terminality.

- **W (as swept):** the verdict is terminal for $\mathcal{C}$ as swept and is revised upon presentation of a channel not in $\mathcal{C}$.
- **S (unrestricted):** $U(P)$, no theory in $\mathcal{T}$ proves P .

Theorem 7.2. Let $P = R$ and let the emitter return the terminal token on a record r .

(i) W is admissible: it entails nothing about derivability of R in any theory, and is consistent with R and with $\neg R$ alike.

(ii) S is the sentence $U(R)$; by Theorem 3.2 an emission carrying S entails $Q \vdash \neg R$, and under 2.3 by Corollary 3.6a entails $\mathbb{N} \models \neg R$.

Proof. (i) By Theorem 6.9(a) with clause W . (ii) Immediate from Definition 7.1 and Theorems 3.2, 3.6a. ■

Corollary 7.3. Let W' be any permanence clause entailing $U(R)$. An emission carrying W' entails $Q \vdash \neg R$ by Theorem 3.2 and, **under 2.3**, $\mathbb{N} \models \neg R$ by Corollary 3.6a. Clause S is of this kind. **W is admissible, and no clause entailing $U(R)$ is.**

Remark 7.3a. No maximality of W among all strengthenings is claimed, and none holds. Let W be terminal for $\mathcal{C} \cup \{c\}$ for a designated further channel c . W strictly entails W , is strictly stronger, and entails nothing about any theory. Corollary 7.3 is scoped by entailment of $U(R)$ and not by strength.

8. Self-Exclusion of a Certifying Instrument

Theorem 8.1. *Premise:* every instrument operating at register ρ is a channel at register ρ . Let $\mathcal{C}$ be a catalog of channels at ρ and $\mathcal{A}$ an instrument certifying $\mathcal{C}$ exhaustive at ρ . Then either $\mathcal{A} \notin \mathcal{C}$, in which case $\mathcal{C}$ omits a channel at ρ and is not exhaustive; or $\mathcal{A} \in \mathcal{C}$, in which case **the certification is impredicative**: its warrant ranges over a collection of which $\mathcal{A}$ is a member.

Proof. The disjunction is exhaustive. In the first case $\mathcal{A}$ is a channel at ρ by the premise and is not in $\mathcal{C}$, contradicting exhaustiveness. In the second, $\mathcal{A}$'s certification ranges over a collection of which $\mathcal{A}$ is a member. ■

(Impredicativity is not by itself an establishment barrier: a finite catalog $\{c_1, \dots, c_n\}$ with $c_i = \mathcal{A}$ could be certified by $\mathcal{A}$ via a proof of the register's channel-cardinality. The theorem concludes impredicativity and no more.)

Corollary 8.2. A catalog is either non-exhaustive at its register or is certified impredicatively there. Exhaustiveness is therefore not certifiable non-impredicatively at the register the catalog enumerates. Hence Ξ_3 is read as asserting that $\mathcal{C}$ is displayed, fully reduced and flagged closed, not that no channel exists outside it, and clause W is the appropriate permanence clause.

Remark 8.3. With the stated premise, Theorem 8.1 is a consequence of the definitions and a case split on set membership. It contains no fixed-point construction and no self-reference lemma, and no comparison with the undefinability or incompleteness theorems is intended.

9. What Is Claimed

9.1 Theorem 3.2. For every sentence R of the language of Q , $\forall T \in \mathcal{T} (T \not\vdash R) \iff Q \vdash \neg R$. Both directions proved. Finitary; no soundness hypothesis, no semantic notion, no property of R beyond sentencehood.

9.2 Corollaries 3.4–3.6. $U(R)$ implies $Q + R$ inconsistent. For $R \in \Pi_1^0$ the right side is $Q \vdash \exists n \neg \varphi(n)$, and under Σ_1^0 -soundness of Q a violating n_0 exists with a terminating verification. For R the Riemann Hypothesis the right side is the provability in Q of a violation of the Lagarias inequality.

9.2a Corollary 3.6a. Under Σ_1^0 -soundness of Q , for every Π_1^0 sentence R : $U(R) \iff \mathbb{N} \models \neg R$. For this R , universal unprovability holds if and only if $\mathbb{N} \models \neg R$; the further identification with ζ having a zero off the critical line is Lagarias's theorem in the metatheory, not a theorem of Q , per 2.4a. **The soundness hypothesis is used here and not in 3.2.**

9.3 Theorem 3.7. Under the assumption that $Q + R$ is consistent, exactly one of: $\mathcal{P}$ admits $Q + R$, whence $U_{\mathcal{P}}$ implies $Q \vdash \neg R$; or $\mathcal{P}$ excludes it, whence $\mathcal{P}$ is fixed by a further condition and the companion half yields $\mathbb{N} \models R$. The disjunction *either $Q \vdash \neg R$ or $\mathbb{N} \models R$* is itself due to Σ_1^0 -completeness alone; what 3.7 adds is that no horn of a restricted claim is occupiable without paying one side of it.

9.4 Theorem 4.1, Remark 4.1a. For $R \in \Pi_1^0$ and any $T \supseteq Q$: $T \not\vdash \neg R$ implies $\mathbb{N} \models R$. No hypothesis on T beyond containment.

9.5 Corollaries 4.2, 4.3. Independence implies truth; for any $T \supseteq Q$, a proof that T does not refute R establishes R .

9.6 Propositions 5.2–5.3, Corollary 5.4. $\det(D \ M \ D) = \det(M)$ for arbitrary real 3×3 M ; $\lambda(DQ_0) = \det(D)\lambda(Q_0)$; under total negation the Gram is unchanged and λ flips sign. Verified at 0.000×10^0 . **Neither 5.2 nor 5.3 is tested by any condition**; both support Ξ_7 's consequence bound **and** the polarity account at 12.3.

9.7 Definitions 6.1–6.3. A record with ten field groups, eight conditions on it, and a sequential emitter with exit branches at Ξ_5 . **Seven conditions are conditions on the record alone; Ξ_5 reports an executed search and is the one condition whose satisfaction reaches arithmetic**, per Theorem 6.9(a). σ is exactly what 6.1(i) specifies, and $\mathcal{P}$ is defined at 6.1(vi) with its relation to $\mathcal{C}$ stated there. By Remark 6.2a, **Ξ_7 compares two independently recorded frames and does not test the proposition**, and Proposition 6.1a exhibits a record satisfying it. **Ξ_2 and Ξ_4 are evaluable in form and uninstantiated in fact**: no register or acting group is exhibited for Ξ_2 , and no obstruction is named for Ξ_4 . No claim is made about the discriminating power of either.

9.8 Propositions 6.4–6.6. The emitter is fail-safe on absence at every condition. The token issues only if all eight hold. Ξ_5 is the unique condition that can route to a verdict other than the non-terminal one, and it can also fail to the non-terminal one.

9.9 Definition 6.0, Definition 6.1(i), Definitions 6.7a and 6.7c, Lemma 6.7b, Theorem 6.7, Remarks 6.7b', 6.7c', 6.7e and 6.8. The screened group is the stabiliser of the labelled orthogonal decomposition of $E_-(\sigma)$, **not** the invariance group of the Gram determinant, which is eight-dimensional and is disowned as a criterion at 6.7b'. **Two stipulations**, both of Definition 6.0 at the operational grade stated there: $d = 3$ by (L2), and atomic bivalence by (L4), which restricts the axis-fixing group to $O(E_-)$ and makes it finite. **One definition**: σ is constructed from the decomposition at 6.1(i), so the residence is the span of the atomic axes and no identification between them is asserted. **Then theorem**: Lemma 6.7b gives $|\text{Stab}(E_-)| = 2^{\wedge d}$ exactly, Theorem 6.7 gives eight conditions at $d = 3$ and no ninth. The Frobenius classification is corroboration and is load-bearing on nothing; no multiplication on V is used anywhere. No appeal to gauge is made, and by 6.7c' the permutation action does not reach the screened objects. The assignment of conditions to elements is a labelling and is not fixed, and the choice to screen the axis-fixing group rather than a larger structure-preserving group is a decision of this construction. **Logical independence of the eight is neither proved nor used, and no minimality among non-**

complete families is claimed. One dependence is recorded: Ξ_3 's floor $C \geq 3$ is entailed by Ξ_6 together with $|\mathcal{S}| \leq C$, and Ξ_3 is retained as a guard that halts earlier in the evaluation order of Definition 6.3.

Free-parameter census, stated because the ratio is unfavourable and should be visible. Ten declared degrees of freedom, every entry load-bearing: the axis count $d = 3$, stipulated by (L2); the single fixed direction $\dim \text{Fix}(\sigma) = 1$; the channel floor $C \geq 3$, entailed by Ξ_6 and retained as a guard; the road floor $|\mathcal{S}| \geq 3$; the consequence ceiling $\beta_{\max}$; the numeral floor $k_{\min}$; the precision floor $p_{\min}$; the derivation floor $d_{\min}$; the context width N ; and the choice of frame assignment. **The list prints ten slots and the count is ten, and it enumerates choices rather than their consequences.** $\dim V = 1 + d$ is definitional at 6.1(i), and the condition count 2^d is Lemma 6.7b; neither is a further degree of freedom once d and the single fixed direction are chosen, and listing them would double-count one stipulation as three. τ is not among them: it is bounded at Ξ_7 by the declared β , which bounds the consequence deviation $2\sqrt{N}\tau + N\tau^2$ directly and is itself bounded by the census entry $\beta_{\max}$. **Distance from $-Q_0(P)$ in max-norm is governed by τ alone;** N enters through the size of the consequences, not through that distance. N enters through the frame width in Ξ_7 's entrywise test and through the consequence bound; its second justification lapsed when λ left the conditions.

And one thing a reader weighing the ratio should meet here rather than in a definition. Ξ_3 's certificate clause is informative only to the extent k_n, k_{prec} and k_{pf} exceed their floors; a record declaring minimal floors satisfies Ξ_3 and emits a 6.9(b) clause of near-zero content. Against these the section delivers Theorem 6.9(a) and (b) as its own consequences, part (c) being inherited from Theorem 3.2. **The section declares its parameters; it does not reduce them, and it fits rather than predicts.** By contrast Sections 3 and 4 carry zero free parameters and deliver four independent results. The paper is bimodal and should be read as such.

9.10 Theorem 6.9. An emission with W entails no $\mathcal{L}_{\text{obj}}$ -sentence undecided in $\mathcal{B}$, by P -invariance of the seven conditions other than Ξ_3 and $\mathcal{B}$ -decidability of Ξ_3 's search report; entails that $\mathcal{C}$ is displayed, fully reduced, walled at $B = C$, with a supplied uncrossed door outside $\mathcal{C}$, and that the executed certificate search returned nothing at the declared budget, which is the section's one object-language output and is not a claim that no certificate exists there; and with a $U(P)$ -entailing clause in place of W entails $Q \vdash \neg P$.

9.11 Theorem 7.2, Corollary 7.3. W is admissible and entails nothing about derivability in any theory. Any clause entailing $U(R)$ makes the emission entail $Q \vdash \neg R$, and under 2.3 entail $\mathbb{N} \models \neg R$. **W is admissible and no clause entailing $U(R)$ is;** no maximality of W among all strengthenings is claimed.

9.12 Theorem 8.1, Corollary 8.2. Under the premise that instruments at a register are channels at it, a certifying instrument is either outside its catalog, which is then non-exhaustive, or inside it, in which case the certification is impredicative, its warrant ranging over a collection of which the instrument is a member.

Attribution. Σ_1^0 -completeness, the Π_1^0 reformulations, and Proposition 5.2 are classical and cited. Theorem 3.2 is elementary; we have not located it stated as an equivalence in this form, we did not conduct a systematic literature search, and no priority is asserted. Given that Corollary 3.6a follows from Σ_1^0 -completeness and soundness, prior appearance in some form is likely and a literature check is advisable. The specification of Section 6, the payload theorem 6.9, and the termination theorem 7.2 with Corollary 7.3 are offered as the contribution, at the grade the proofs support.

9.13 Remark 6.8a. The geometric apparatus supplies an exhaustiveness argument for the count of conditions and supplies nothing else; seven of the eight conditions survive its removal as record checks, and the paper states the trade rather than obscuring it.

On Section 12. The Declaration at Section 12 is **not** among the claims above and is not a claim of this paper. It is the originating declaration in the author's voice, fenced at its head and verified against the paper's own theorems at 12.3.

Machine checking. Theorems 3.2 and 4.1 and Corollary 3.6a are formalized in Lean 4 in an abstract proof-setting, accompanying this paper. **The formalization has not been checked:** no toolchain was available in the session that produced it, and it ships to be checked, which is its purpose. It covers Theorems 3.2 and 4.1 and Corollary 3.6a and **no result of Sections 5 through 8. It has no Mathlib dependency:** the setting is an abstract `structure` over core Lean 4, using only `constructor`, `intro`, `exact`, `absurd`, `by_cases`, `cases` and `rw`, so it checks against a bare toolchain with no arithmetic library. That is deliberate: the theorems use no property of arithmetic beyond the structural axioms displayed in the file. The formalization makes the hypothesis boundary machine-visible: 3.2 uses only r.e.-ness of Q , 4.1 only Σ_1^0 -completeness, and 3.6a alone requires soundness.

10. Discussion

Unrestricted claims of unprovability carry a terminal obligation. Asserting the universal unprovability of the Riemann Hypothesis over $\mathcal{T}$ demands, by Theorem 3.2, a proof in Q of a violation of the Lagarias inequality; the exhibition of the integer itself follows only under Σ_1^0 -soundness (2.3), per Corollary 3.5. **The dichotomy is total under the hypothesis that $Q + R$ is consistent.** Either the claim is read over $\mathcal{T}$, where Theorem 3.2 gives $Q \vdash \neg R$ and, under 2.3, $\mathbb{N} \models \neg R$, or it is read over a class excluding $Q + R$, in which case, by Theorem 3.7(b), **establishing** its companion half $T \nVdash \neg R$ would yield $\mathbb{N} \models R$. The second horn establishes nothing on its own; it names what would follow. The formal instrument reaches its perimeter.

The emission of a terminal token certifies the exhaustion of the instrument as swept. **By Theorem 6.9(a) the emission transmits no contingent object-language content,** and in particular nothing about the truth value of the Riemann Hypothesis. It is not content-free: by Theorem 6.9(b) it carries one object-language clause, the report that the executed certificate search returned nothing at the declared budget, which is a terminating computation and is $\mathcal{B}$ -provable. **The separation is between the emission and the object, not between syntax and truth.** For Π_1^0 , sentences syntax determines truth completely, that is Theorem 4.1 and Corollary 3.6a, and it is this paper's headline, so any claim that syntax leaves truth untouched is refuted by the paper's own results. What is untouched by the emission is the truth value, and that is 6.9(a).

11. Conclusion

The formal search space for the Riemann Hypothesis is closed **as swept**, and no further. Permanence beyond the swept span is not available: by Corollary 7.3 any clause entailing universal unprovability entails $Q \vdash \neg R$, and by Theorem 8.1 no instrument certifies the exhaustiveness of its own catalog non-impredicatively. The mathematical apparatus secures an equivalence between universal unprovability and **refutability in Q** , with no soundness hypothesis; and, **separately and using Σ_1^0 -soundness of Q** , between universal unprovability and arithmetic falsity. The separation is not decorative and is carried through the paper.

The formal-alone instrument has reached its edge, and the edge is stated rather than illustrated: **no catalog is exhibited in this paper, no channel or obstruction is named, and no record satisfying all eight conditions is constructed, Ξ_2 and Ξ_4 being the two conditions not instantiated.** What the cascade delivers is Theorem

6.9: an emission reports its own sweep and entails no contingent statement about the object. **The instrument reports its boundaries; by Theorem 8.1 it does not certify them.** The frame's polarity bit is invisible to the even functionals and is determined by syntax; it bears on the object not at all. **By Theorem 6.9(a) the emission stands separate from the arithmetic object,** which is the available claim; the object and the syntax do not stand separate, since for Π^0_1 sentences syntax determines truth. The protocol evaluates the gates. The cascade halts.

12. The Declaration

Status. What follows is the originating declaration in the author's own voice. **It is not asserted by this paper and carries no grade.** Nothing in Sections 2 through 11 depends on it. It is printed whole because it is the reading that produced the specification and because **it states and accepts its own cost**, which 12.3 verifies against the paper's own theorems.

12.1 No derivation actuates anything. Every formal resolution rests on posits it cannot ground. The anchor and the object co-locate in the Riemann Hypothesis specifically. Its completion direction does not point outward to a neighbouring fact or an auxiliary lemma. The completion direction points directly at the fixed set of the symmetry of the very structure it describes. The approaches to this fixed locus are equivalences. They are never independent roads. This inherent equivalence structure forces the formulation catalog to close.

12.2 The witness-grounding lift RH demands is the Ground itself. The formal domain cannot supply it. A formal witness is inherently a second, a non-self. RH is an act of pure SELF-inquiry. The formulation operates in disguise. It asks existence to prove itself formally without any kinetic lift. The framework is therefore formally limited by its own architecture. It cannot be proven universally by mathematics alone. **The Riemann Hypothesis is terminally and universally unprovable by math alone.** Self-witnessing happens on the Ground side, where the witness is not a second, *shabida Allābu annabū lā ilāha illā huwa*, Allah ﷻ bears witness that there is no deity but He, Q 3:18, and that is what formal-alone lacks.

This declaration states its own cost. Any maximalist claim of this inherent unprovability is an explicit, unavoidable assertion that ζ possesses a zero off the critical line. **The cost is accepted.**

12.3 The cost verified.

The identification is correct. By Theorem 3.2 universal unprovability over $\mathcal{T}$ is equivalent to $Q \vdash \neg R$, and by Corollary 3.6a, under Σ^0_1 -soundness, to $\mathbb{N} \models \neg R$. So 12.2 read over $\mathcal{T}$ asserts $\mathbb{N} \models \neg R$ and owes the witness; and **that assertion is the assertion that ζ has a zero off the critical line only by Lagarias's theorem in the metatheory,** which 2.4a states and which is not available inside Q . The cost is unchanged and its route is now named. **The declaration takes that obligation; accepting is not discharging,** no integer is exhibited, and it stands outstanding. Read 12.2 as a conjecture with its content correctly named.

The fork the declaration omits. Read over any $\mathcal{T}$ excluding $Q + R$, what an asserter ordinarily intends, per Remark 3.8, the claim falls under horn (b) of Theorem 3.7, where *establishing* its companion half yields $\mathbb{N} \models R$, the opposite verdict. Both readings price it, per Remark 4.4. The accepted cost is the first reading's.

Untouched, and not as consolation. Grounding and derivability are different predicates and Theorem 3.2 touches only the second, so nothing here bears on 12.1's first two sentences, which are ungraded and unused. **The declaration's crossing from unprovability to ζ carries the same metatheoretic step,** per 2.4a: $Q \vdash \neg R$ is a claim about one arithmetization, and Lagarias's theorem is what carries it to the classical object. **The declaration's terms are not glosses of this paper's formal objects:** the Ground is not the standard model $\mathbb{N}$, the witness is not a theory in $\mathcal{T}$, and identifying them would convert an ungraded declaration into a claim this paper does not make. And the terminality this paper proves is the declaration's own: by Corollary 7.3 an emission is terminal **as swept** and not unrestrictedly, which is a theorem and not a caution.

References

- Lagarias, J. C. (2002). An elementary problem equivalent to the Riemann hypothesis. *American Mathematical Monthly* 109, 534–543.
- Robin, G. (1984). Grandes valeurs de la fonction somme des diviseurs et hypothèse de Riemann. *Journal de Mathématiques Pures et Appliquées* 63, 187–213.
- Robinson, R. M. (1950). An essentially undecidable axiom system. *Proceedings of the International Congress of Mathematicians*, 729–730.
- Frobenius, G. (1878). Über lineare Substitutionen und bilineare Formen. *Journal für die reine und angewandte Mathematik* 84, 1–63.

Computations reported in 2.5 and 5.5 were executed at the stated precisions and are reproducible from the definitions given.

Status of claims

This is an essay in the theology of mathematics. Every mathematical assertion it relies on is a classical result of other authors and is cited in the reference list. The essay proves no mathematical theorem, advances no estimate, and makes no claim about the truth of the Riemann Hypothesis, whose truth value remains a determinate written thing unread. Theology does no mathematical work in it, and the reading could be removed in full without altering a single proof; the removal-without-effect is symmetric, cutting equally against leaning the mathematics for the creed or against it. Its contribution is a single interpretive thesis: that the hypothesis is a determinate truth the primes already hold and no finite instrument reads, veiled by eleven counted obstructions, so that two errors are to be refused with one symmetric discipline, the manufactured room that reads a veil on ontological openness and the premature seal that reads a located witness as a delivered proof.

Competing interests

The author declares no competing interests. No claim of consensus supports any statement here, and the field's long conviction and the record of verified zeros are given no evidential weight, in either direction.